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Electronics Basic

6.2

Bipolar Junction Transistors



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Bipolar Junction Transistors

Topics Covered in Chapter 6

6-1: Transistor Construction

6-2: Proper Transistor Biasing

6-3: Operating Regions

6-4: Transistor Ratings

6-5: Checking a Transistor with an Ohmmeter

6-6: Transistor Biasing



6-1: Transistor Construction

- A transistor has three doped regions, as shown in Fig. 6-1 (next slide).
- Fig. 6-1 (a) shows an npn transistor, and a pnp is shown in (b).
- For both types, the base is a narrow region sandwiched between the larger collector and emitter regions.



6-1: Transistor Construction

- The emitter region is heavily doped and its job is to emit carriers into the base.
- The base region is very thin and lightly doped.
- Most of the current carriers injected into the base from emitter pass on to the collector.
- The collector region is moderately doped and is the largest of all three regions.

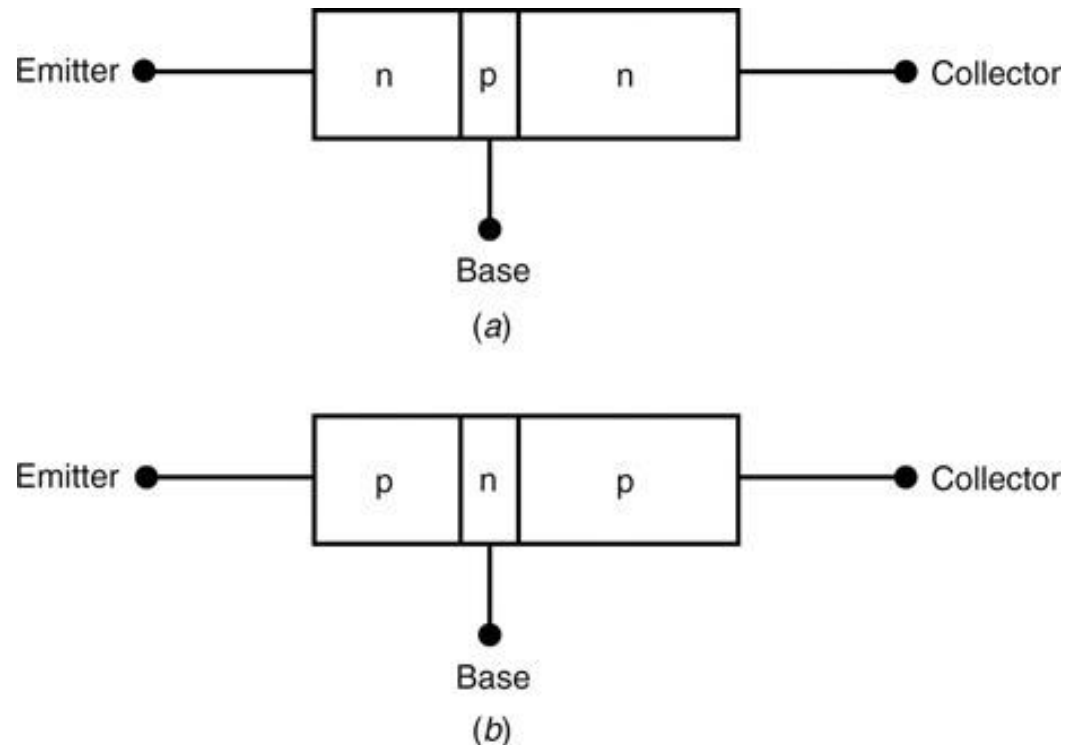
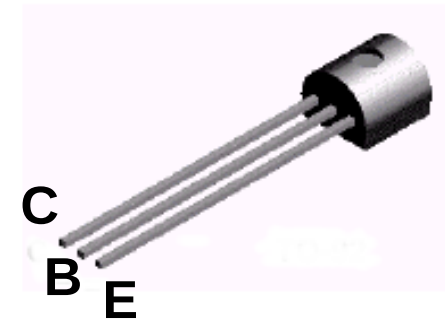
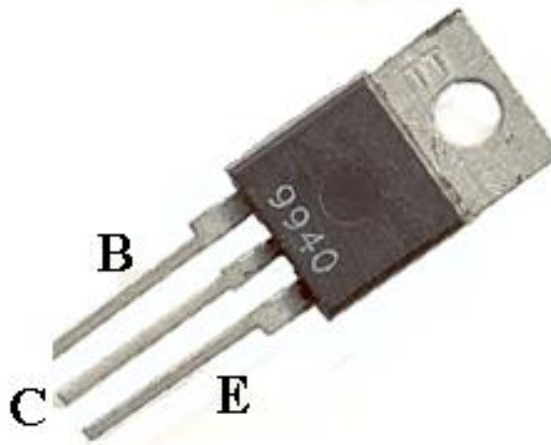
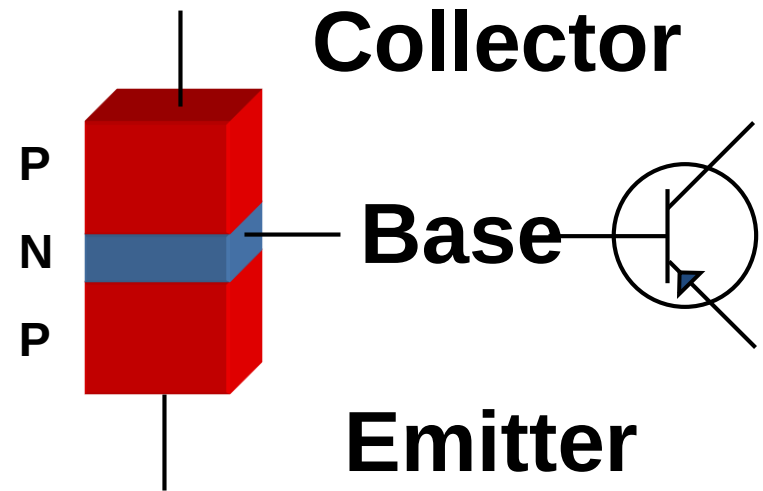
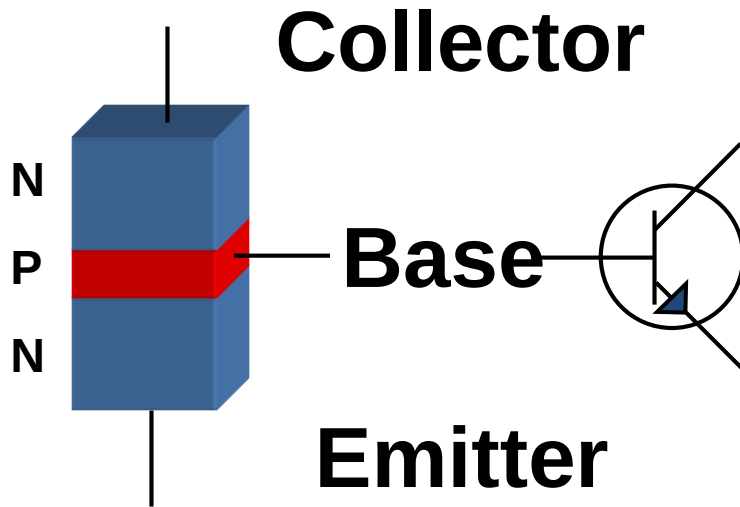


Fig. 6-1



Bipolar Transistors

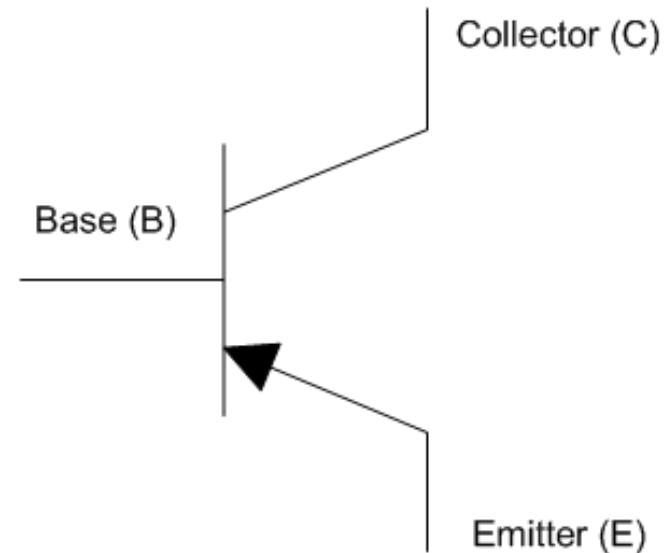
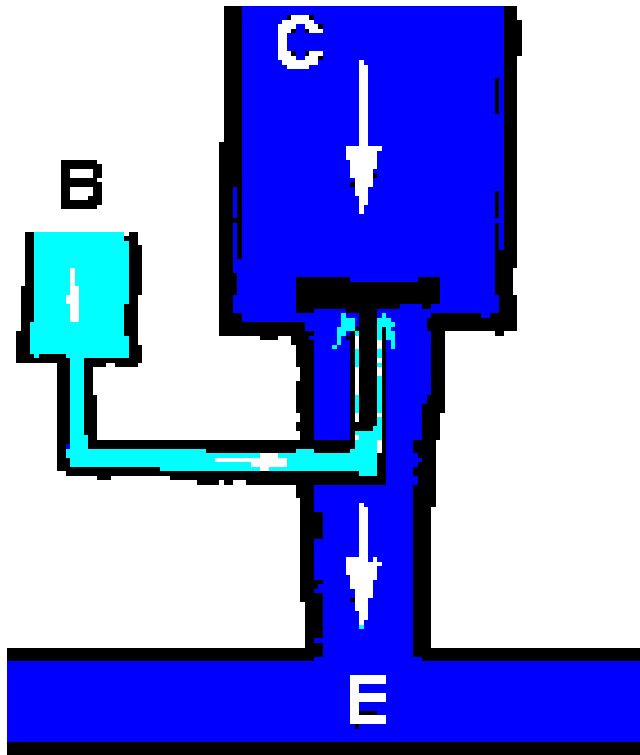


6-2: Proper Transistor Biasing

- For a transistor to function properly as an amplifier, the **emitter-base** junction must be forward-biased and the **collector-base** junction must be reverse-biased.
- The common connection for the voltage sources are at the base lead of the transistor.
- The emitter-base supply voltage is designated V_{EE} and the collector-base supply voltage is designated V_{CC} .
- For silicon, the barrier potential for both EB and CB junctions equals 0.7 V



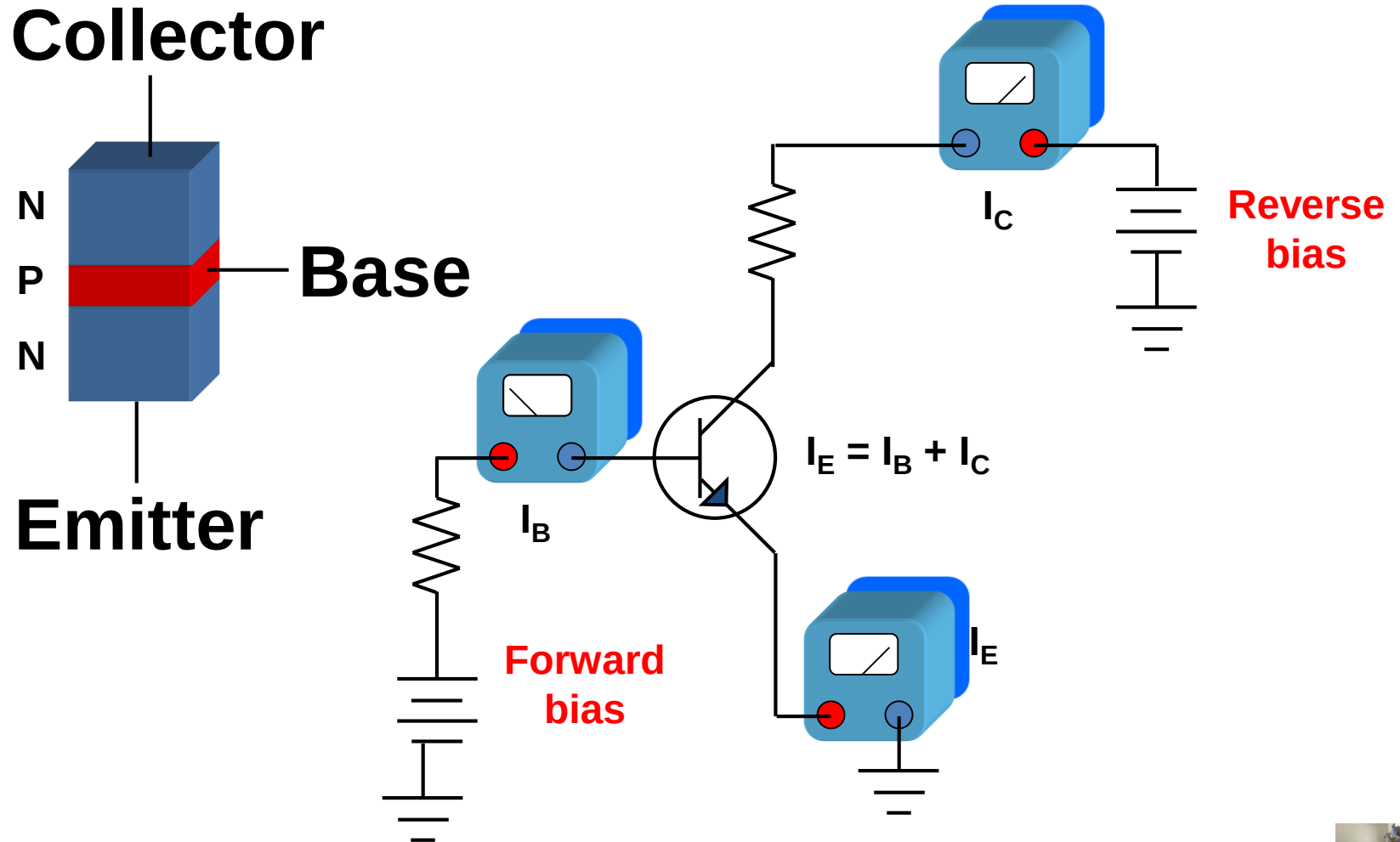
Schematic Symbol



PNP transistor



Transistor Biasing



6-2: Proper Transistor Biasing

- Fig. 6-4 shows transistor biasing for the *common-base* connection.
- Proper biasing for an npn transistor is shown in (a).
- The EB junction is forward-biased by the emitter supply voltage, V_{EE} .
- V_{CC} reverse-biases the CB junction.
- Fig. 6-4 (b) illustrates currents in a transistor.
- CE voltage of an npn transistor must be positive
- Ratio of I_C to I_E is called DC alpha α_{dc}

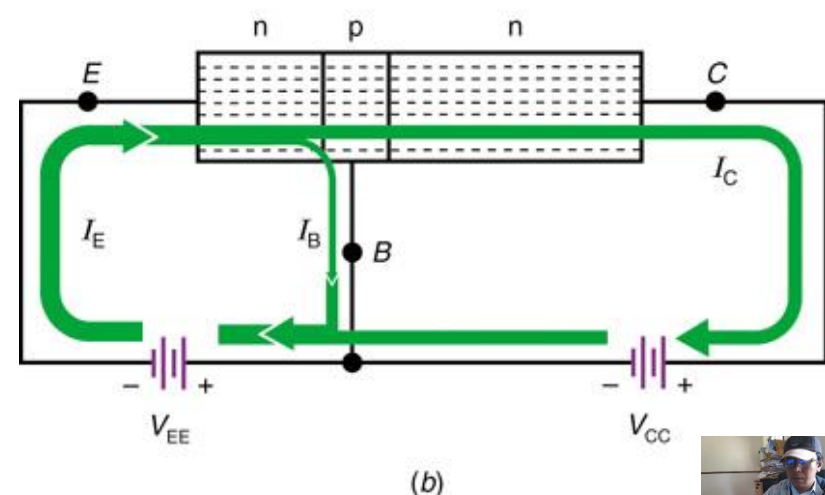
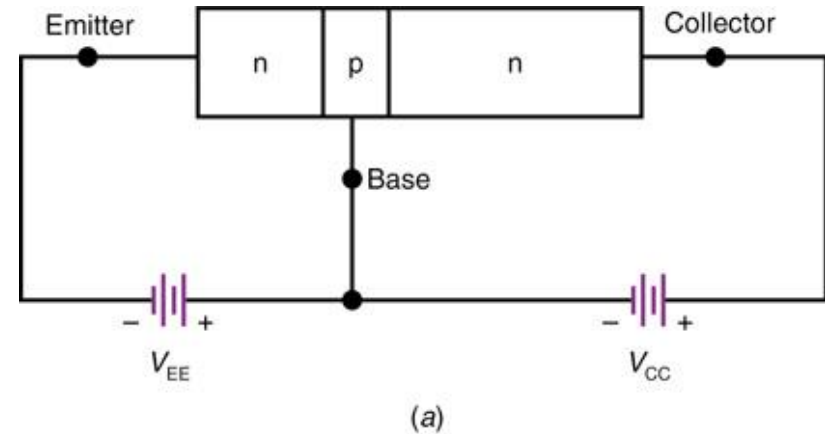


Fig. 6-4



6-3: Operating Regions

- Since emitter lead is common, this connection is called common-emitter connection
- Collector current I_C is controlled solely by the base current, I_B .
- By varying I_B , a transistor can be made to operate in any one of the following regions
 - Active
 - Saturation
 - Breakdown
 - Cutoff
- Ratio of I_C to I_B is called DC beta β_{dc}

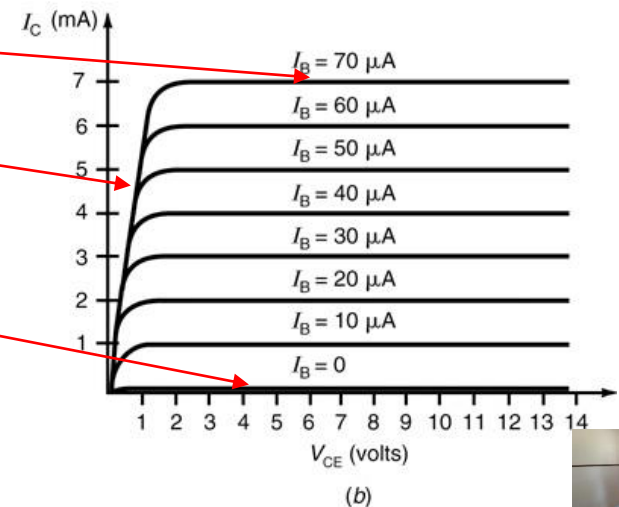
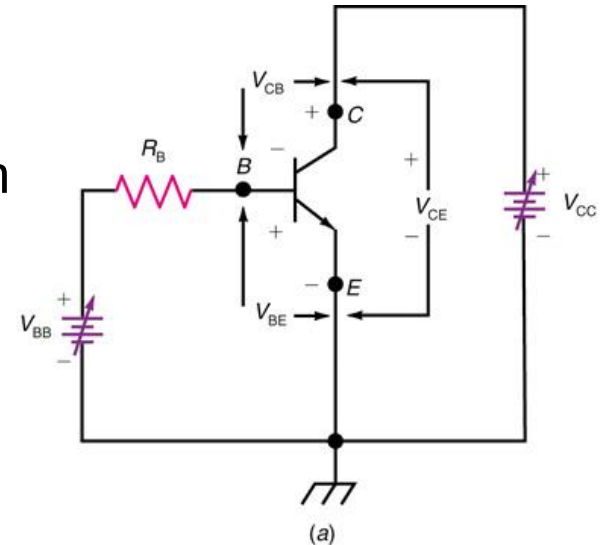


Fig. 6-6: Common-emitter connection (a) circuit.
(b) Graph of I_C versus V_{CE} for different base current values.



6-3: Operating Regions

- Active Region
 - Collector curves are nearly horizontal
 - I_C is greater than I_B ($I_C = \beta_{dc} \times I_B$)
- Saturation
 - I_C is not controlled by I_B
 - Vertical portion of the curve near the origin
- Breakdown
 - Collector-base voltage is too large and collector-base diode breaks down
 - Undesired collector current
- Cutoff
 - $I_B = 0$
 - Small collector current flows $I_C \approx 0$



Transistor Currents

- $I_E = I_B + I_C$
- $I_C = I_E - I_B$
- $I_B = I_E - I_C$
- $\beta_{dc} = \frac{I_C}{I_B}$
- $\alpha_{dc} = \frac{I_C}{I_E}$
- $\alpha_{dc} = \frac{\beta_{dc}}{1 + \beta_{dc}}$



Example 6-4

- A transistor has the following currents:

$$I_E = 15 \text{ mA}$$

$$I_B = 60 \text{ } \mu\text{A}$$

Calculate α_{dc} , and β_{dc}

- $I_C = I_E - I_B = 14.94 \text{ mA}$

- $\alpha_{dc} = 0.996$

- $\beta_{dc} = 249$



6-4: Transistor Ratings

- A transistor, like any other device, has limitations on its operations.
- These limitations are specified in the manufacturer's data sheet.
- Maximum ratings are given for
 - Collector-base voltage
 - Collector-emitter voltage
 - Emitter-base voltage
 - Collector current
 - Power dissipation



6-5: Checking a Transistor with an Ohmmeter

- An analog ohmmeter can be used to check a transistor because the emitter-base and collector-base junctions are p-n junctions.
- This is illustrated in Fig. 6-8 where the npn transistor is replaced by its diode equivalent circuit.

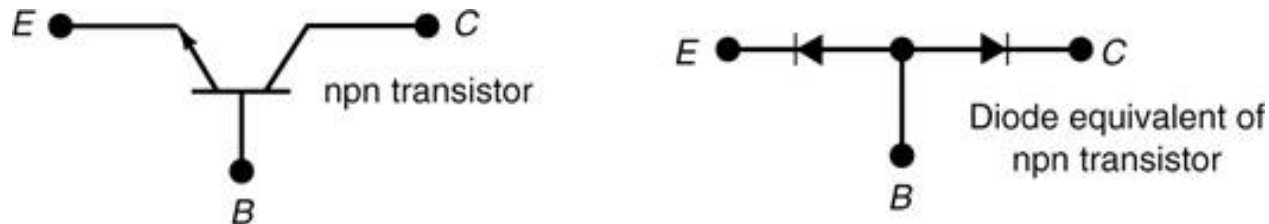


Fig. 6-8



Using a DMM to check a Diode

- Ohmmeter ranges in DMMs do not provide the proper forward bias to turn on the diode
- Set DMM to the special diode range
- In forward-bias, digital display indicates the forward voltage dropped across the diode
- In reverse-bias, digital display indicates an over range condition
- For silicon diode, using an analog meter, the ratio of reverse resistance, R_R , to forward resistance, R_F , should be very large such as 1000:1 or more



6-5: Checking a Transistor with an Ohmmeter

- To check the base-emitter junction of an npn transistor, first connect the ohmmeter as shown in Fig. 6-9 (a) and then reverse the ohmmeter leads as shown in (b).
- For a good p-n junction made of silicon, the ratio R_R/R_F should be equal to or greater than 1000:1.

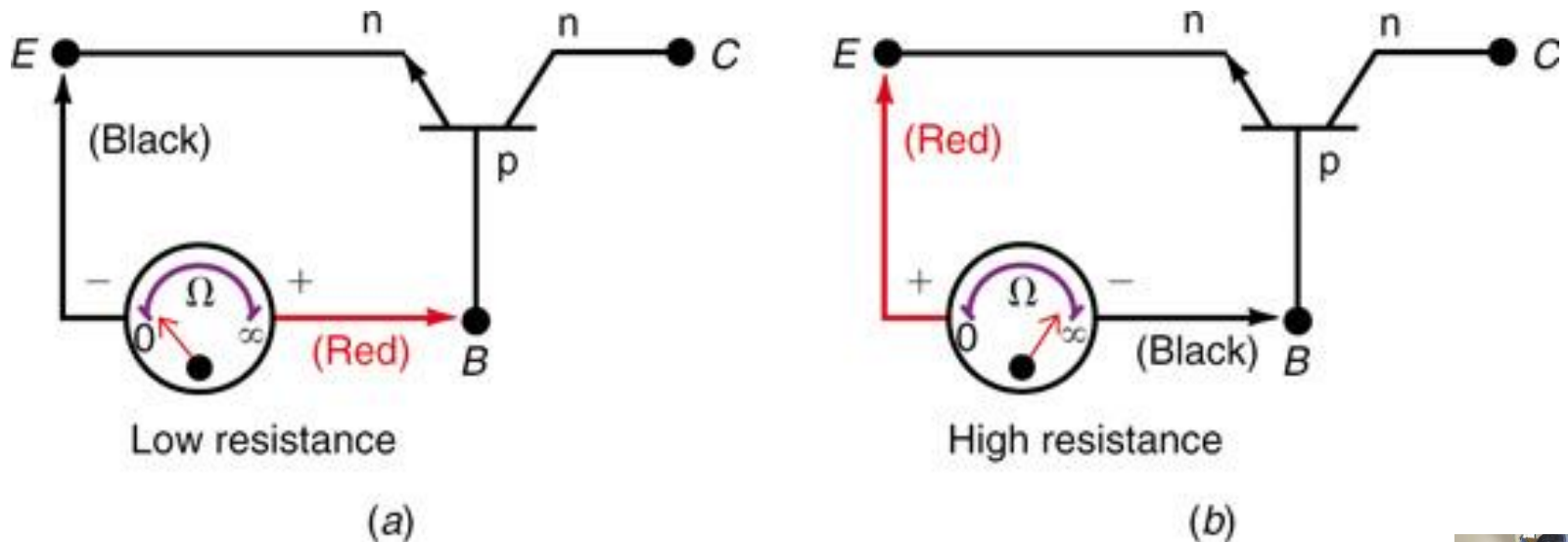


Fig. 6-9



6-5: Checking a Transistor with an Ohmmeter

- To check the collector-base junction, first connect the ohmmeter as shown in Fig. 6-10 (a) and then reverse the ohmmeter leads as shown in (b).
- For a good p-n junction made of silicon, the ratio R_R/R_F should be equal to or greater than 1000:1.
- Although not shown, the resistance measured between the collector and emitter should read high or infinite for both connections of the meter leads.

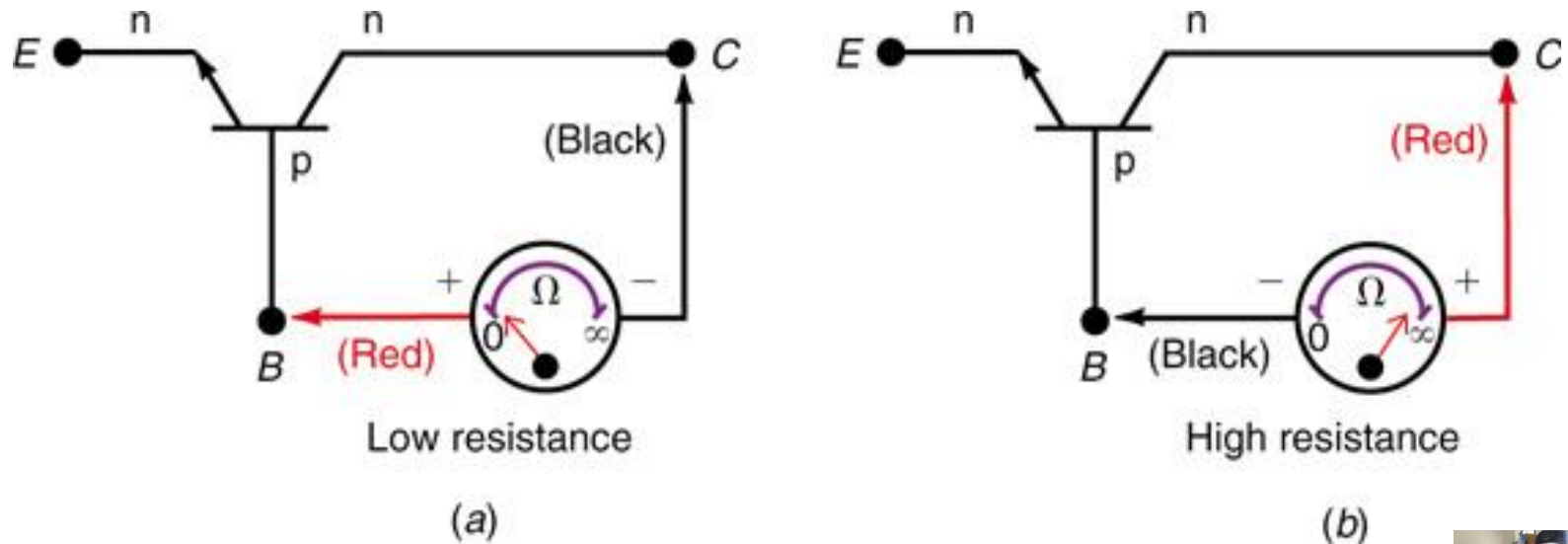


Fig. 6-10



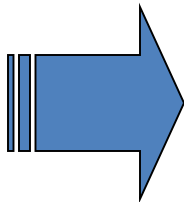
Circuit with BJTs

Our approach: Operating point - dc operating point
Analysis of the signals - the signals to be amplified

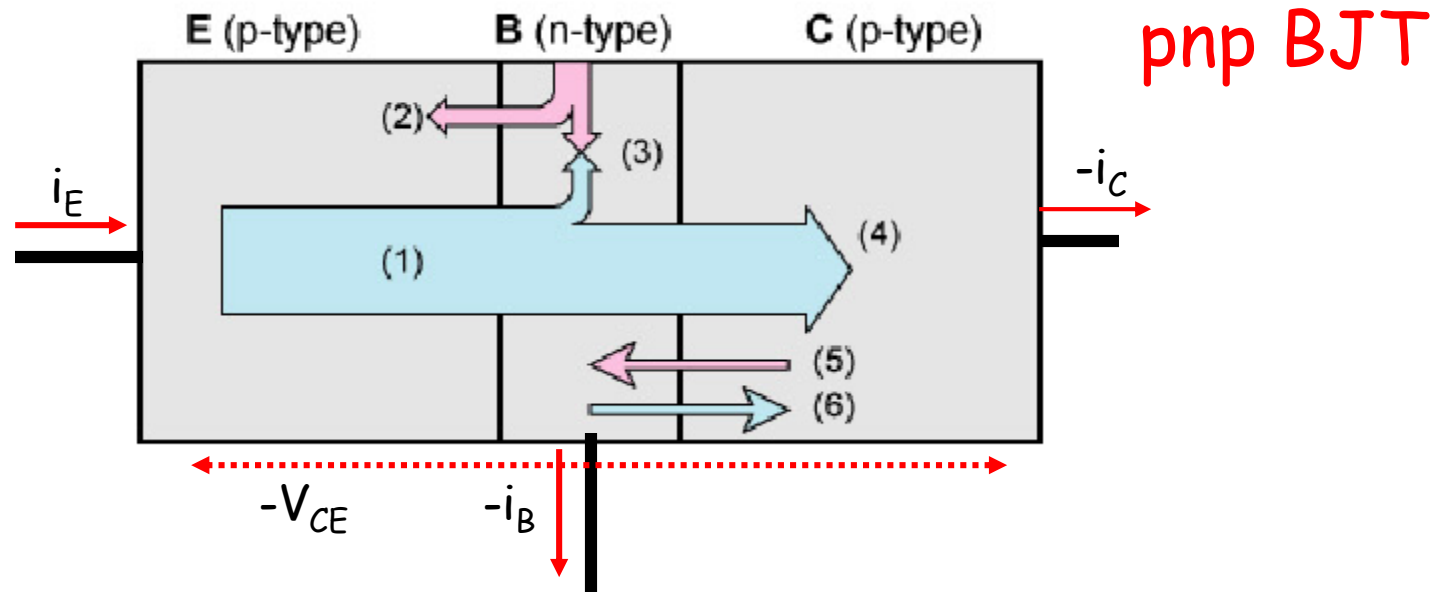
Circuit is divided into: model for large-signal dc analysis of BJT circuit
bias circuits for BJT amplifier
small-signal models used to analyze circuits for signals being amplified

Table 2.3 The Four Regions of Operation of a Bipolar Junction Transistor		
Region of Operation	Emitter-Base Junction	Collector-Base Junction
Forward active	forward biased	reverse biased
Reverse active	reverse biased	forward biased
Cutoff	reverse biased	reverse biased
Saturated	forward biased	forward biased

Remember !



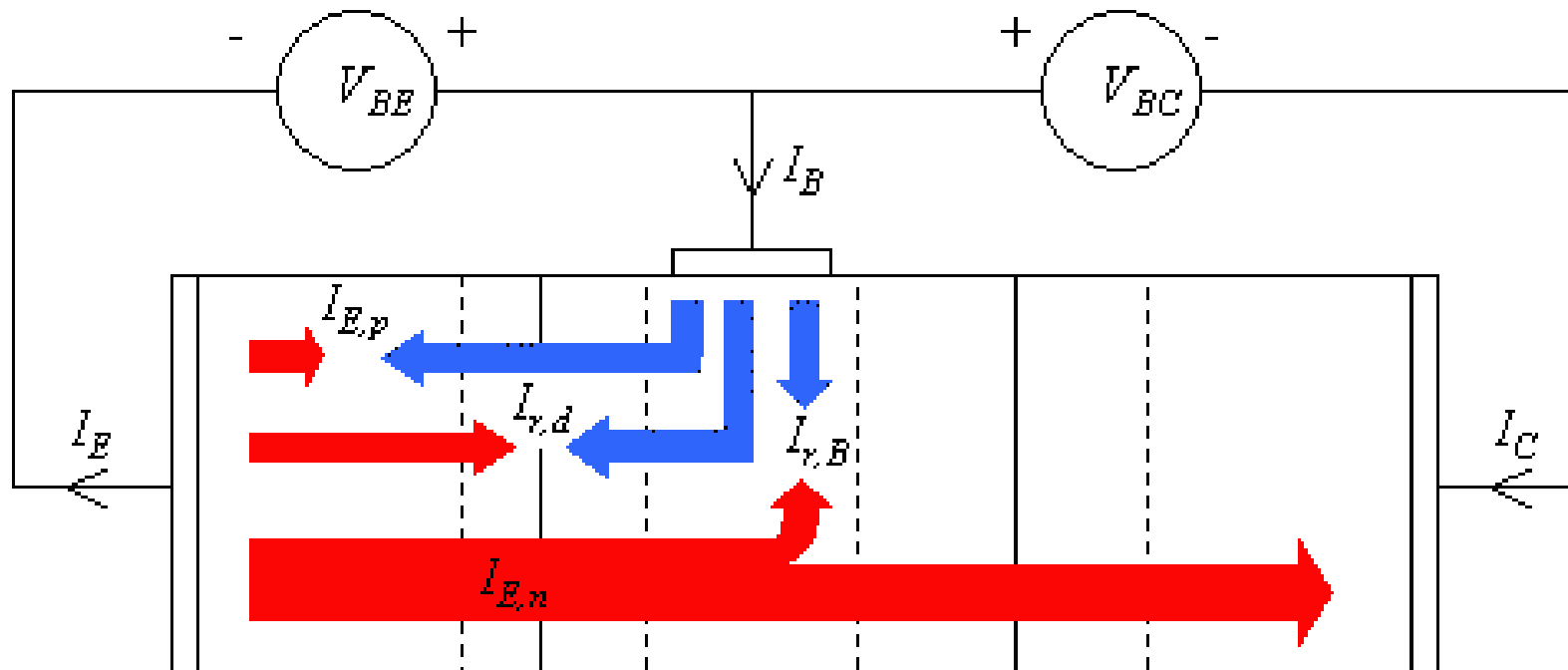
Current Flow in BJT



1. Injected h^+ current from E to B
2. e^- injected across the forward-biased EB junction (current from B to E)
3. e^- supplied by the B contact for recombination with h^+ (recombination current)
4. h^+ reaching the reverse-biased C junction
- 5,6. Thermally generated e^- & h^+ making up the reverse saturation current of the C junction



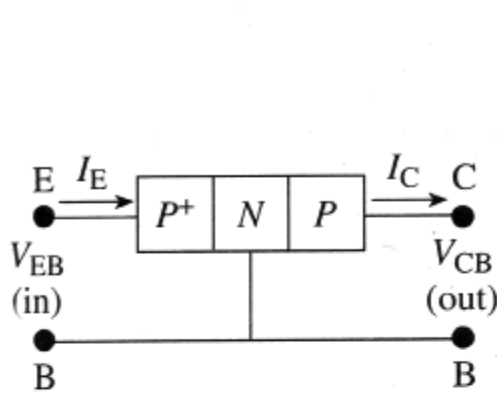
Now, you can try...



npn BJT

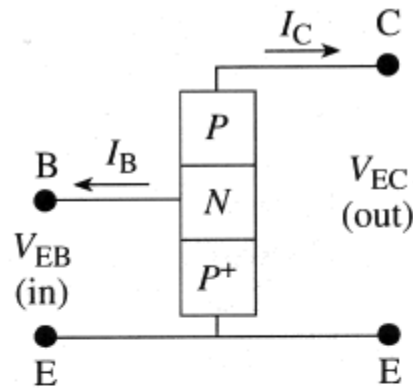


BJTs - Basic configurations



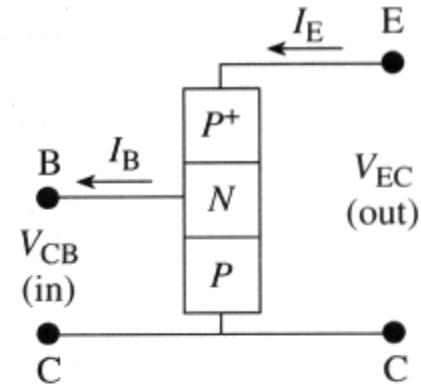
Common base

Both the input and output share the base “in common”



Common emitter

Both the input and output share the emitter “in common”

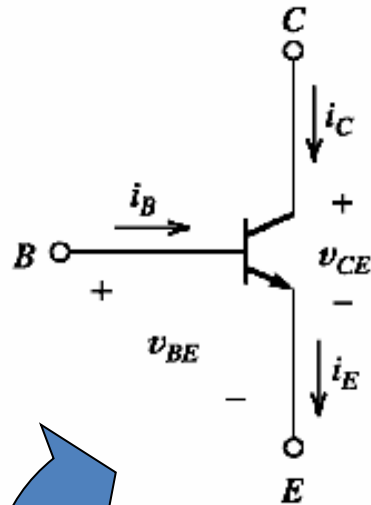


Common collector

Both the input and output share the Collector “in common”



nnpn BJTs - Operation Modes



Forward & reverse polarized
pn junctions

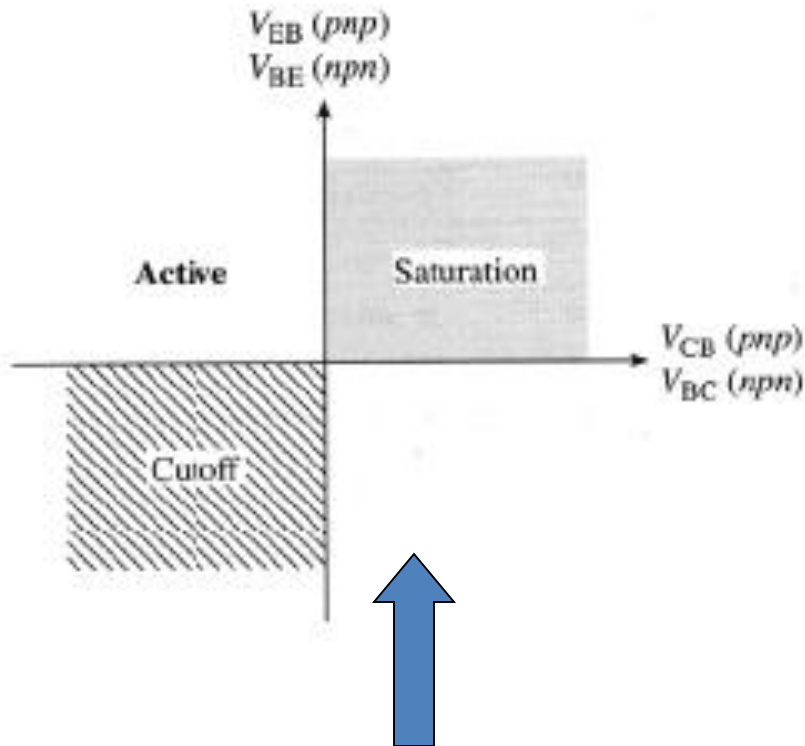
Different operation modes:

Table 2.3 The Four Regions of Operation of a Bipolar Junction Transistor

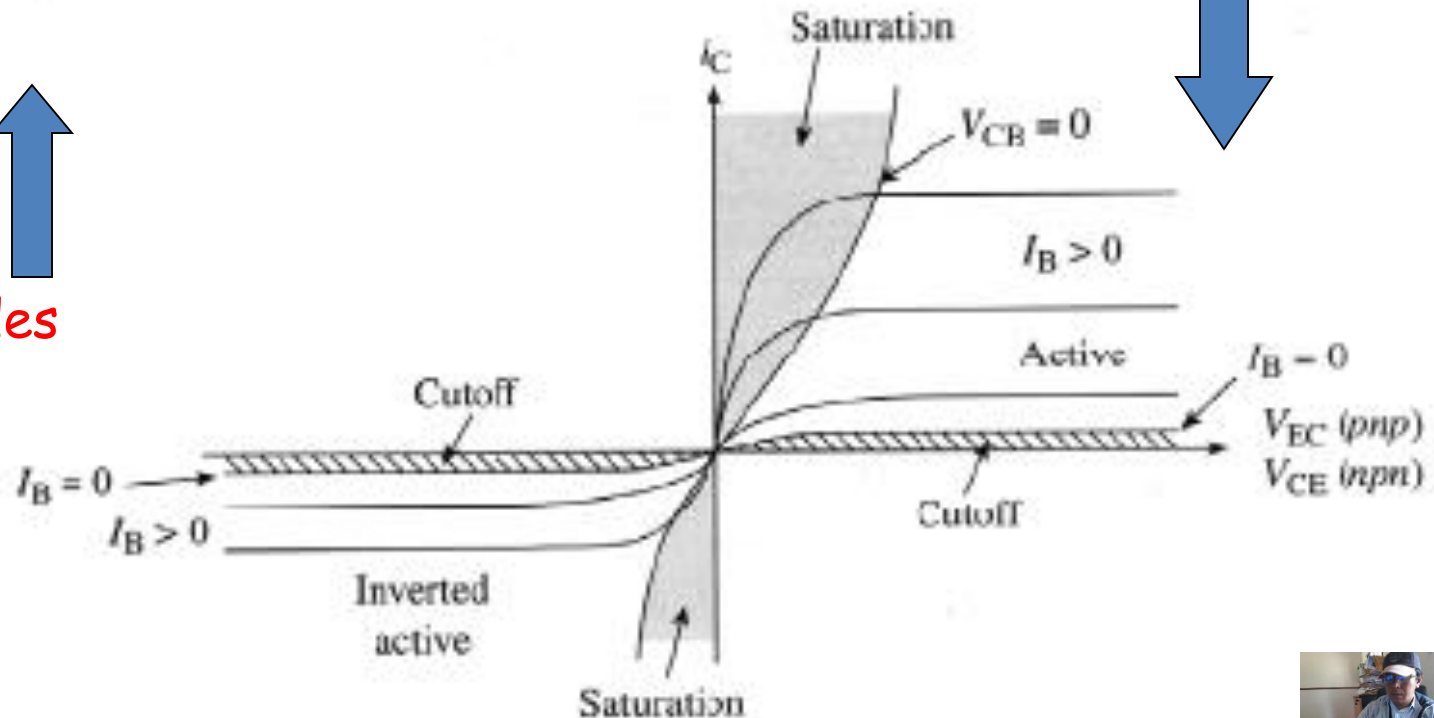
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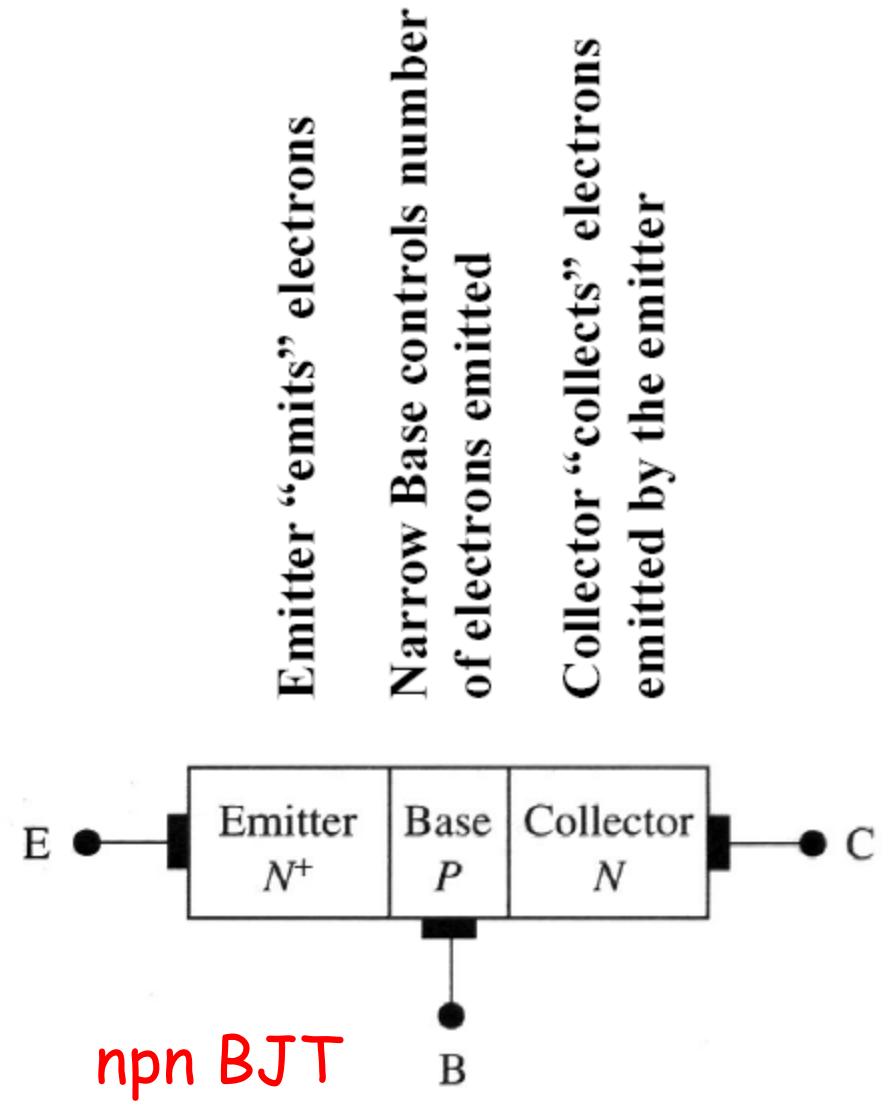
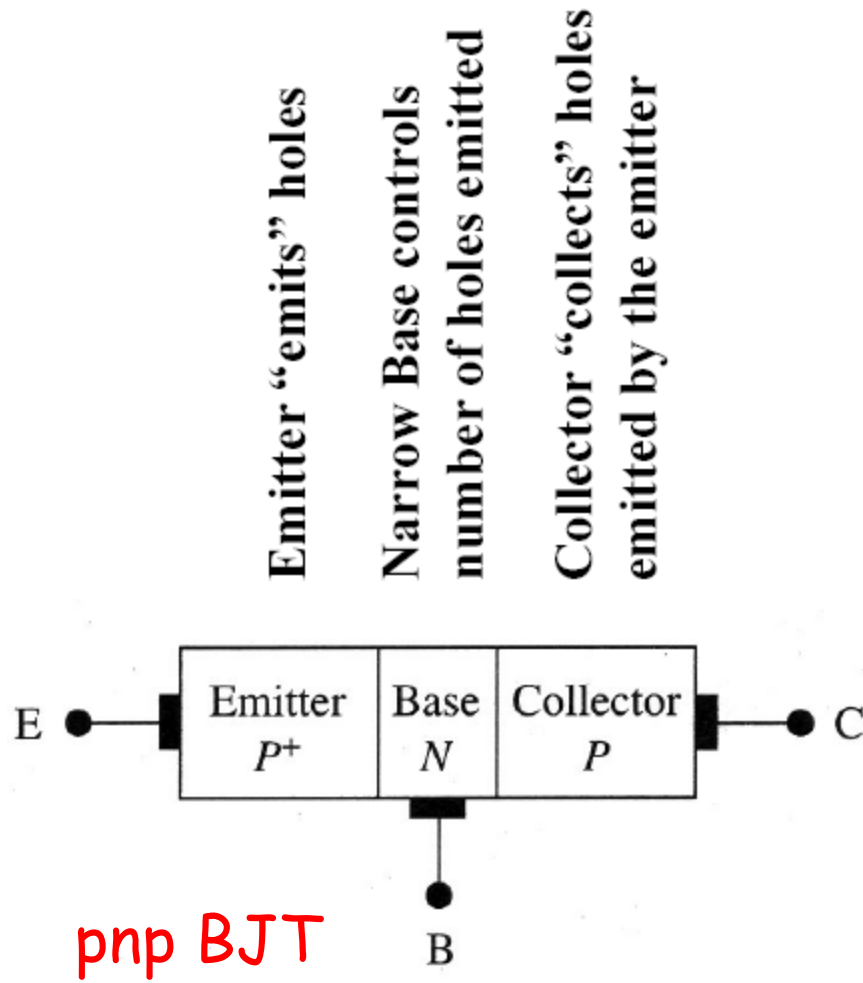
nnp BJTs - Operation Modes



- When there is no I_B current almost no I_C flows
- When I_B current flows, I_C can flow
- The device is then a current controlled current device



BJT-Basic operation



(n^+) , (p^+) - heavy doped regions; Doping in $E > B > C$



BJTs - Current & Voltage Relationships

Operation mode: v_{BE} is forward & v_{BC} is reverse

The Shockley equation

$$i_E = I_{ES} \left[\exp\left(\frac{v_{BE}}{V_T}\right) - 1 \right]$$

Einstein relation

$$\frac{D}{\mu} = \frac{kT}{q}$$

I_{ES} -saturation I (10^{-12} - 10^{-16} A); $V_T = kT/q$ -thermal V (26meV)

D - diffusion coefficient [cm^2/s]
 μ - carrier mobility [cm^2/Vs]

The Kirchhoff's laws

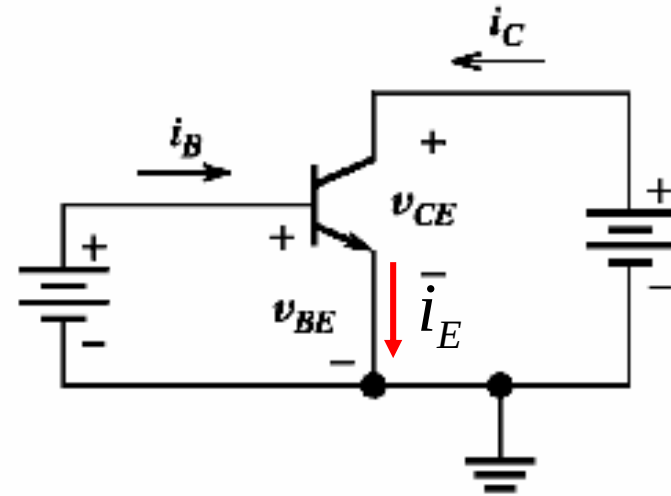
$$i_E = i_C + i_B$$
$$V_{BE} + V_{BC} + V_{CE} = 0$$

It is true regardless of the bias conditions of the junction

Useful parameter

$$\beta = \frac{i_C}{i_B}$$

the common-emitter current gain for ideal BJT β is infinite



BJTs - Current & Voltage Relationships

Useful
parameter

$$\alpha = \frac{i_C}{i_E}$$

the common-base current gain
for typical BJT α is ~ 0.99

The Shockley equation
once more

$$i_C = \alpha I_{ES} \left[\exp\left(\frac{v_{BE}}{V_T}\right) - 1 \right]$$

If we define the scale current

$$I_S = \alpha I_{ES}$$

$$i_C \cong I_S \left(\frac{v_{BE}}{V_T} \right)$$

A little bit of math... search for i_B

$$i_B = (1 - \alpha) i_E$$



$$i_B = (1 - \alpha) I_{ES} \left[\exp\left(\frac{v_{BE}}{V_T}\right) - 1 \right]$$

Finally...

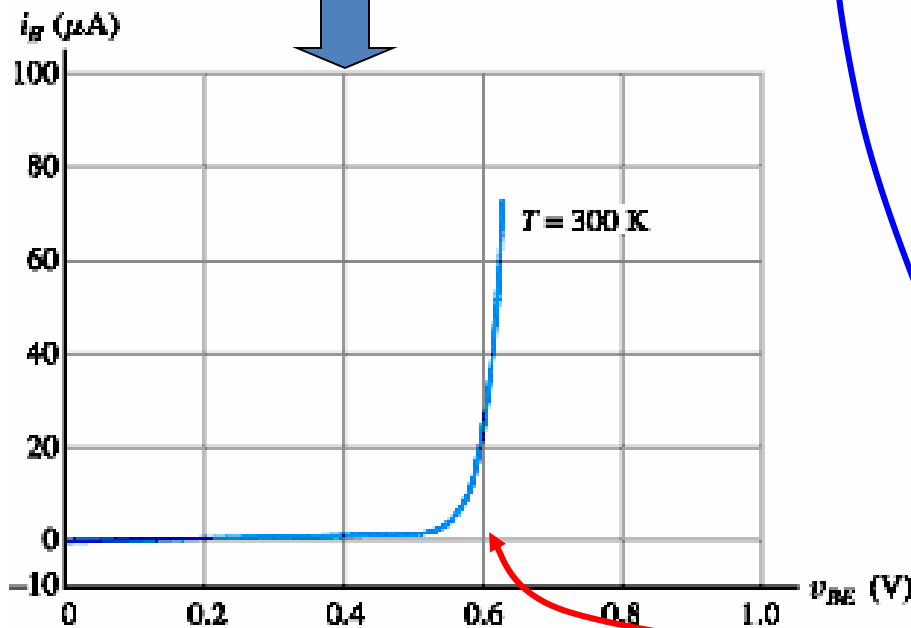
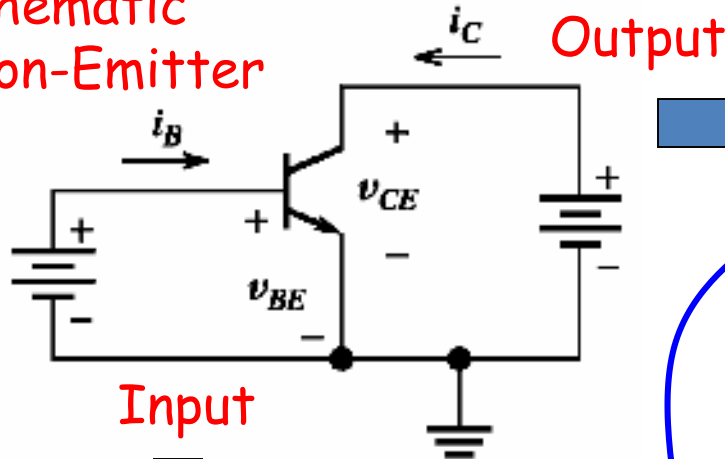
$$\beta = \frac{i_C}{i_B} = \frac{\alpha}{1 - \alpha}$$

$$i_C = \beta i_B$$

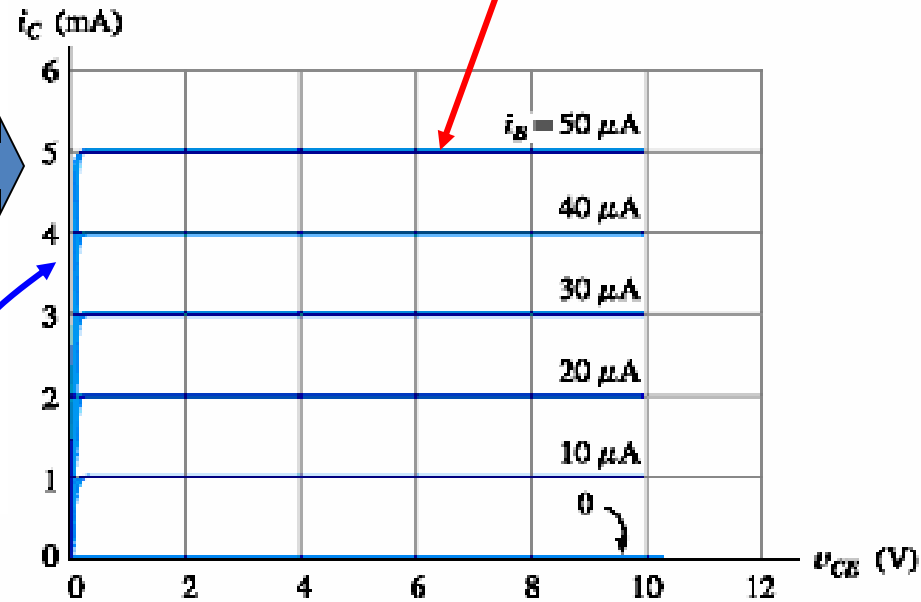


BJTs - Characteristics

Schematic
Common-Emitter



(a) Input characteristic



(b) Output characteristics

$V_{BC} < 0$ or equivalently $V_{CE} > V_{BE}$

If $V_{CE} < V_{BE}$ the B-C junction is forward bias and I_C decreases

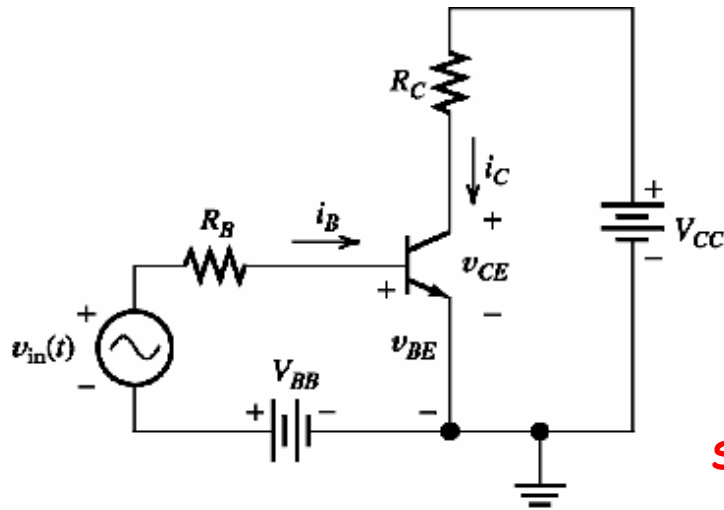
Remember V_{BE} has to be greater than 0.6-0.7 V

Example 13.1



BJTs - Load line analysis

Common-Emitter Amplifier



Input loop

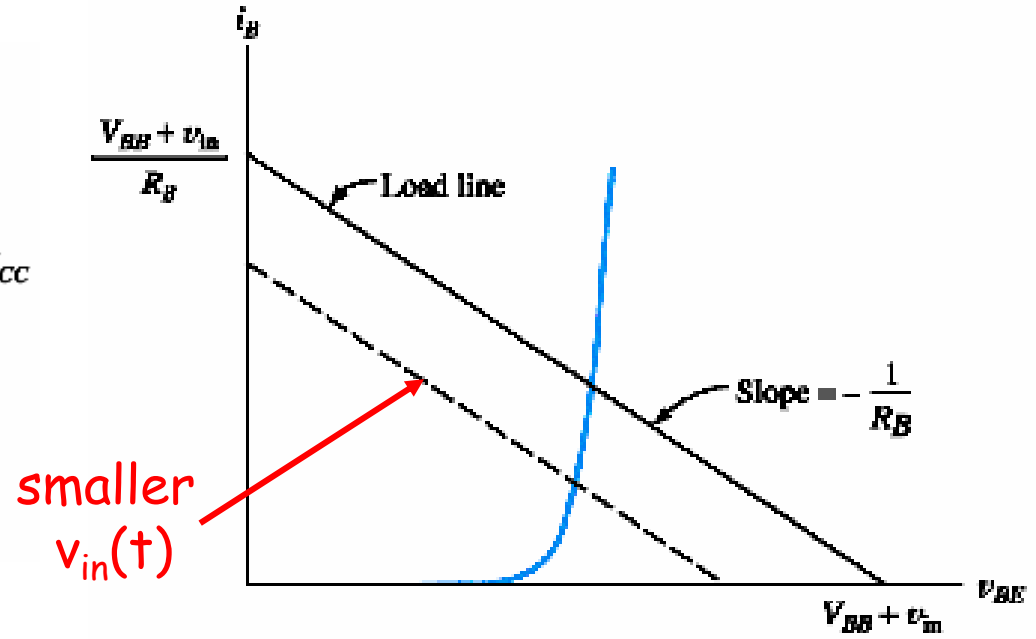
$$V_{BB} + v_{in}(t) = R_B i_B(t) + v_{BE}(t)$$

if $i_B = 0$

$$v_{BE} = V_{BB} + v_{in}$$

if $v_{BE} = 0$

$$i_E = (V_{BB} + v_{in}) / R_B$$

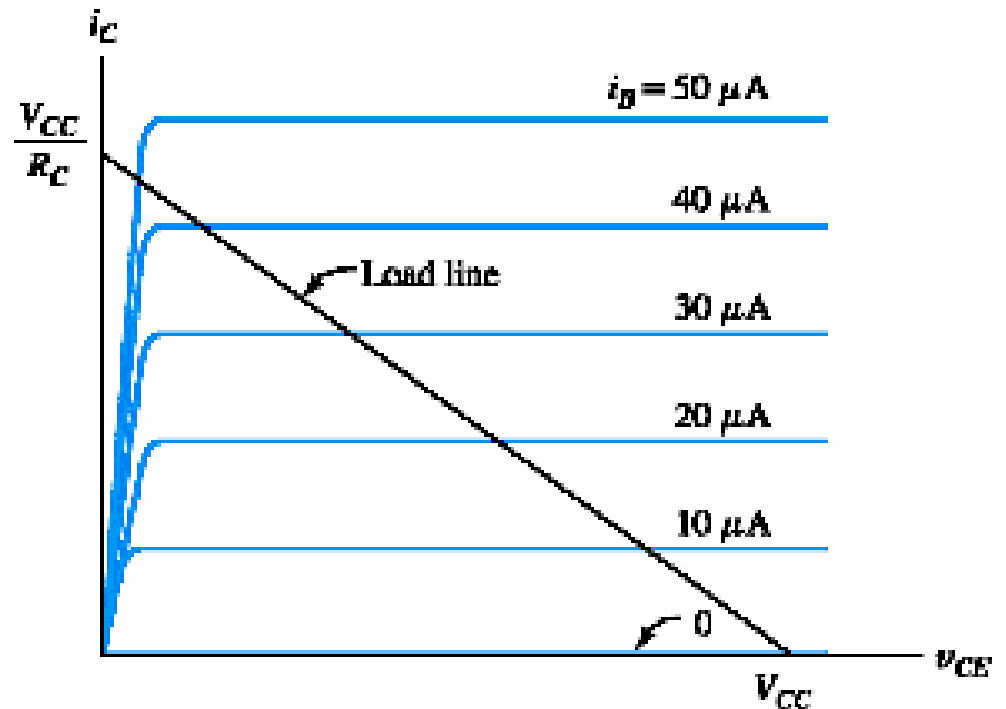
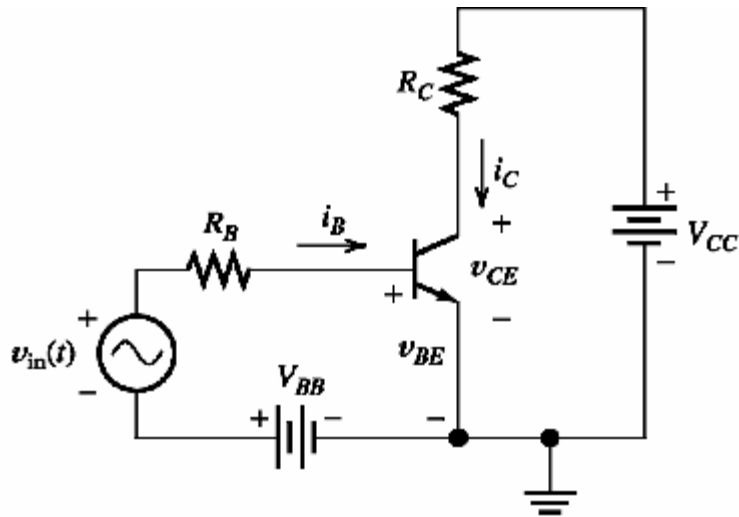


(a) Input load line (shifts to dashed line for a smaller value of v_{in})



BJTs - Load line analysis

Common-Emitter Amplifier



Output loop

$$V_{CC} = R_C i_C + v_{CE}$$

Example 13.2



Circuit with BJTs

Our approach: Operating point - dc operating point

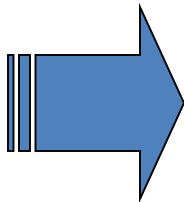
Analysis of the signals - the signals to be amplified

Circuit is divided into: model for large-signal dc analysis of BJT circuit
bias circuits for BJT amplifier
small-signal models used to analyze circuits for signals being amplified

Table 2.3 The Four Regions of Operation of a Bipolar Junction Transistor

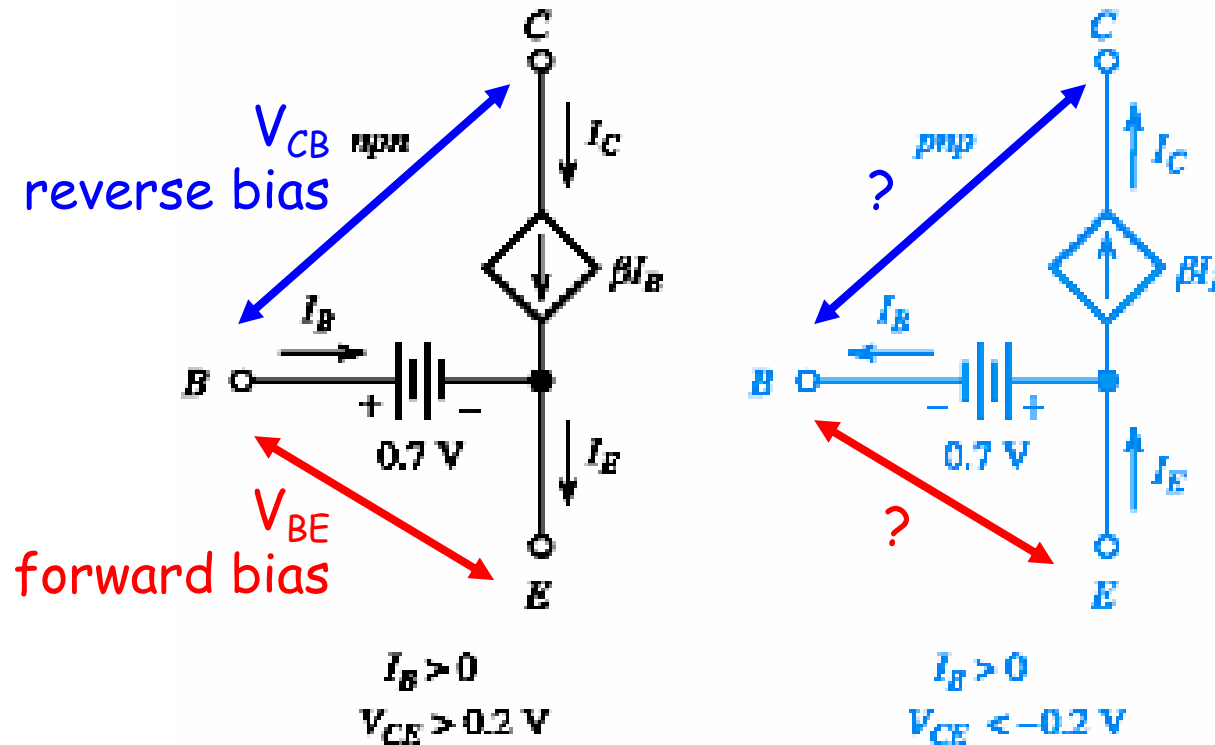
Region of Operation	Emitter-Base Junction	Collector-Base Junction
Forward active	forward biased	reverse biased
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Saturated	forward biased	forward biased

Remember !



Large-Signal dc Analysis: Active-Region Model

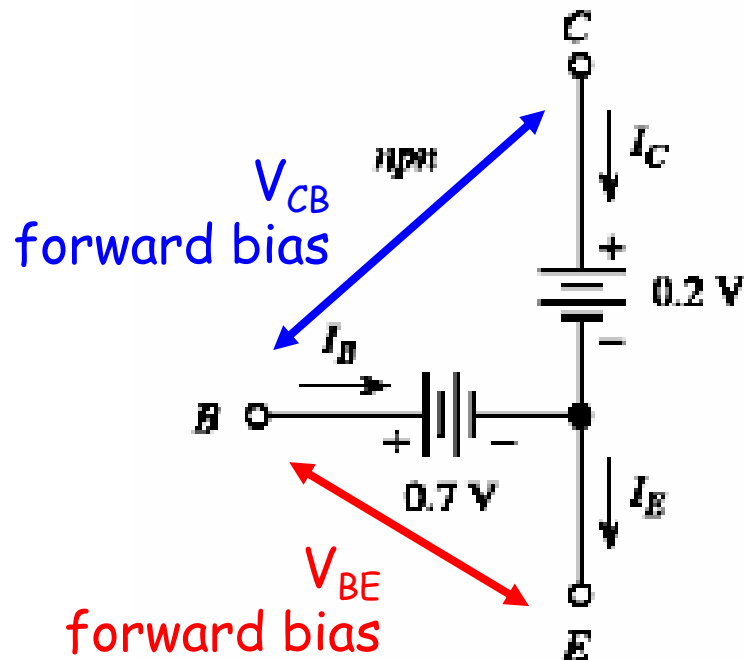
Important: a current-controlled current source models the dependence of the collector current on the base current



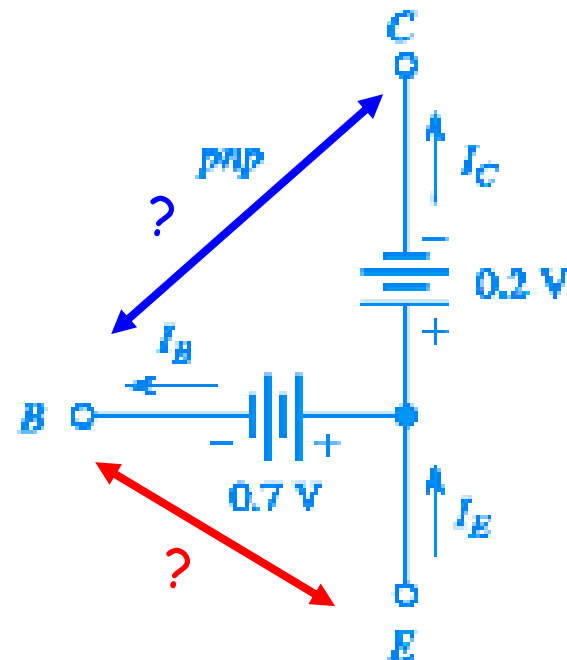
(a) Active region

The constraints for I_B and V_{CE} must be satisfied to keep BJT in the active-mode

Large-Signal dc Analysis: Saturation-Region Model



$$I_B > 0$$
$$\beta I_B > I_C > 0$$

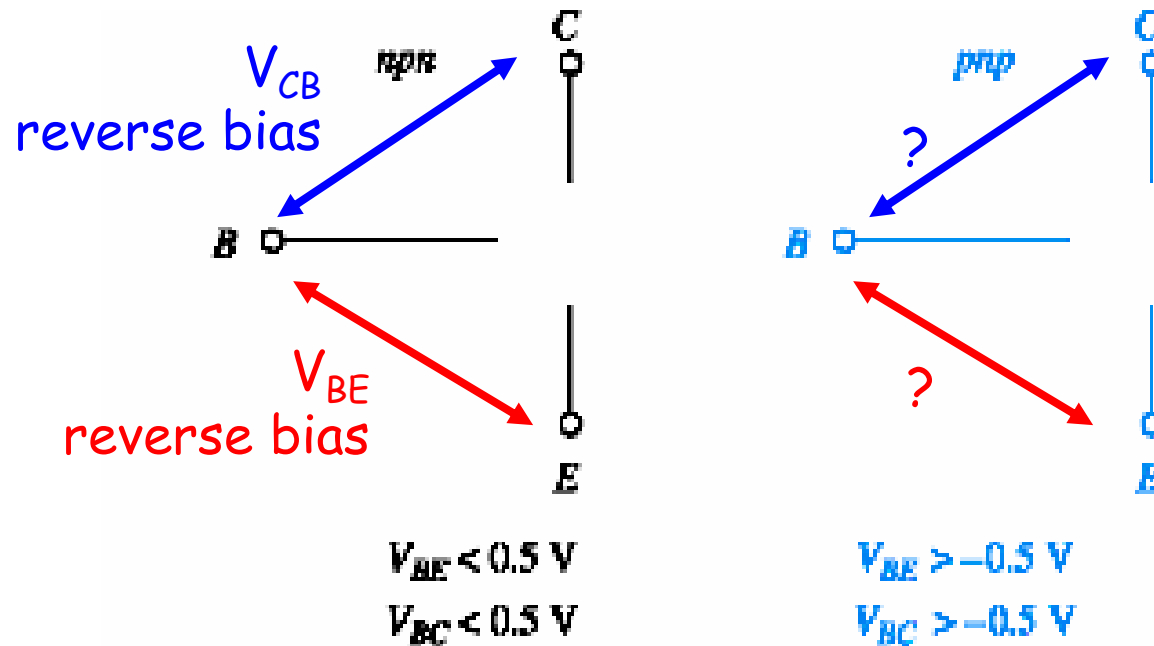


$$I_B > 0$$
$$\beta I_B > I_C > 0$$

(b) Saturation region



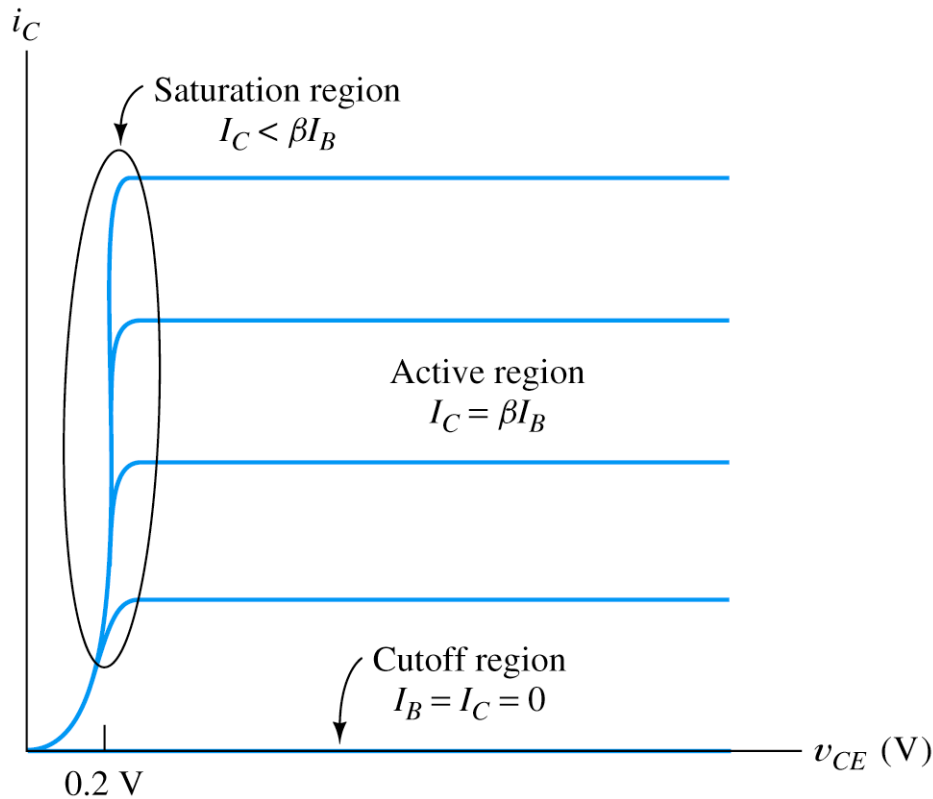
Large-Signal dc Analysis: Cutoff-Region Model



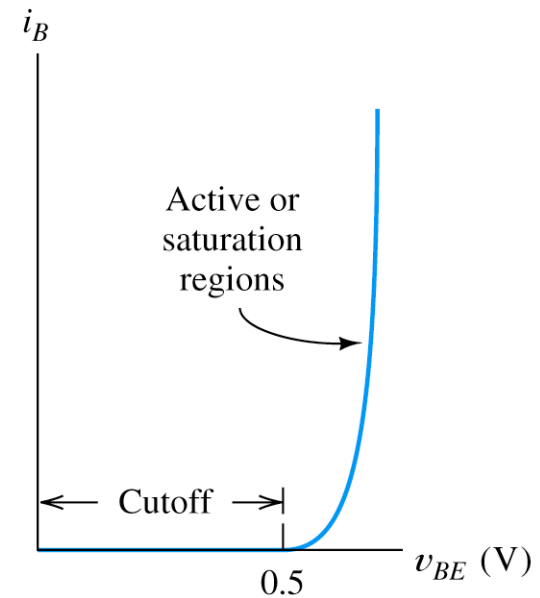
(c) Cutoff region

If small forward-bias voltage of up to about 0.5 V are applied, the currents are often negligible and we use the cutoff-region model.

Large-Signal dc Analysis: characteristics of an npn BJT



(a) Output characteristic



(b) Input characteristic

Figure 13.17 Regions of operation on the characteristics of an *nnp* BJT.



Large-Signal dc Analysis

- Procedure:
- (1) select the operation mode of the BJT
 - (2) use selected model for the device to solve the circuit and determine I_C , I_B , V_{BE} , and V_{CE}
 - (3) check to see if the solution satisfies the constraints for the region, if so the analysis is done
 - (4) if not, assume operation in a different region and repeat until a valid solution is found

This procedure is very important in the analysis and design of the bias circuit for BJT amplifier.

The objective of the bias circuit is to place the operating point in the active region.

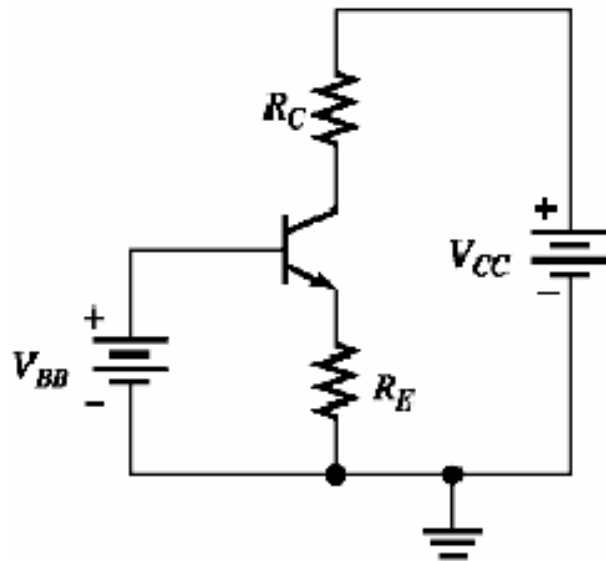
Bias point - it is important to select I_C , I_B , V_{BE} , and V_{CE} independent of the β and operation temperature.

Example 6.4, 6.5, 6.6

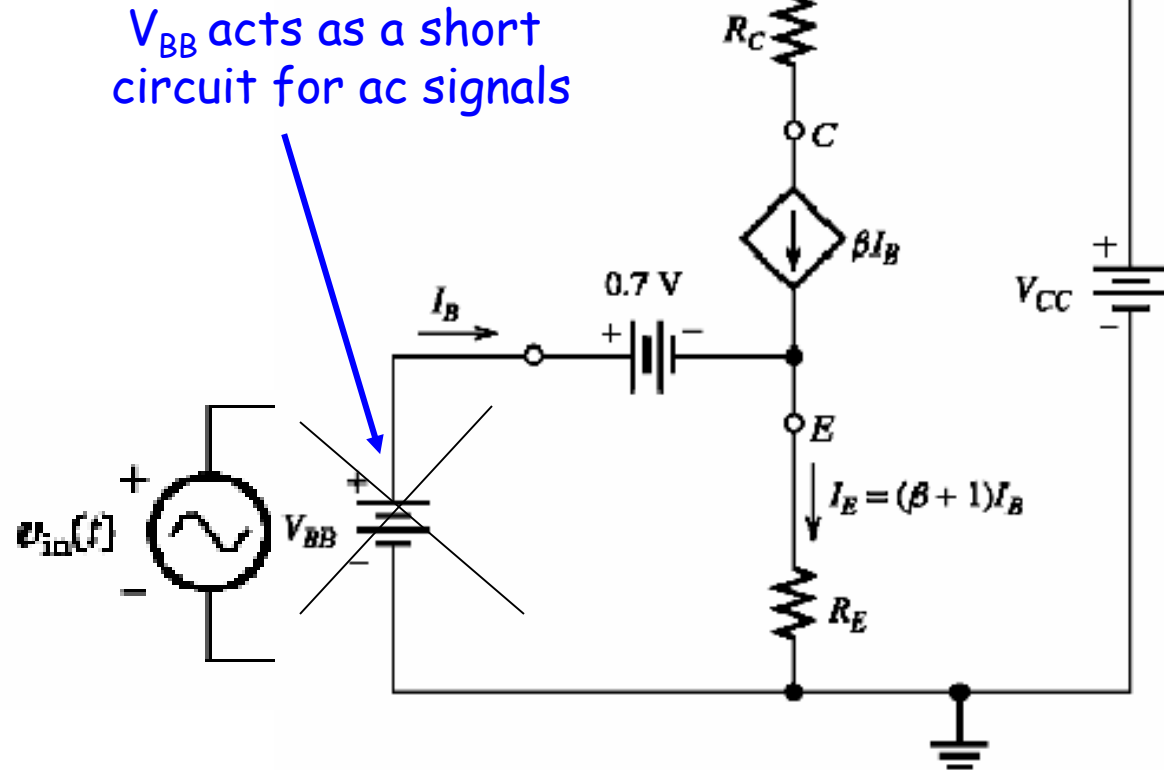


Large-Signal dc Analysis: Bias Circuit

From Example 6.6



(a) Original circuit



(b) Equivalent circuit assuming operation in the active region

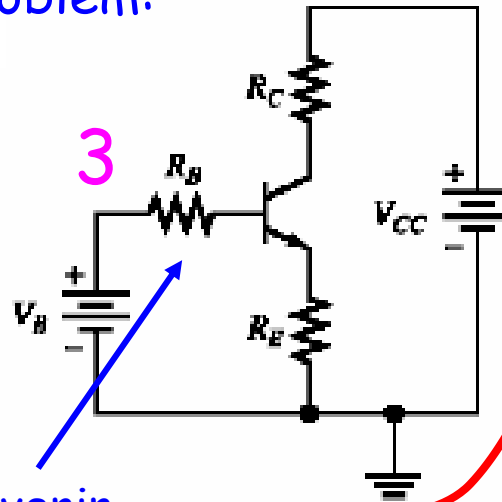
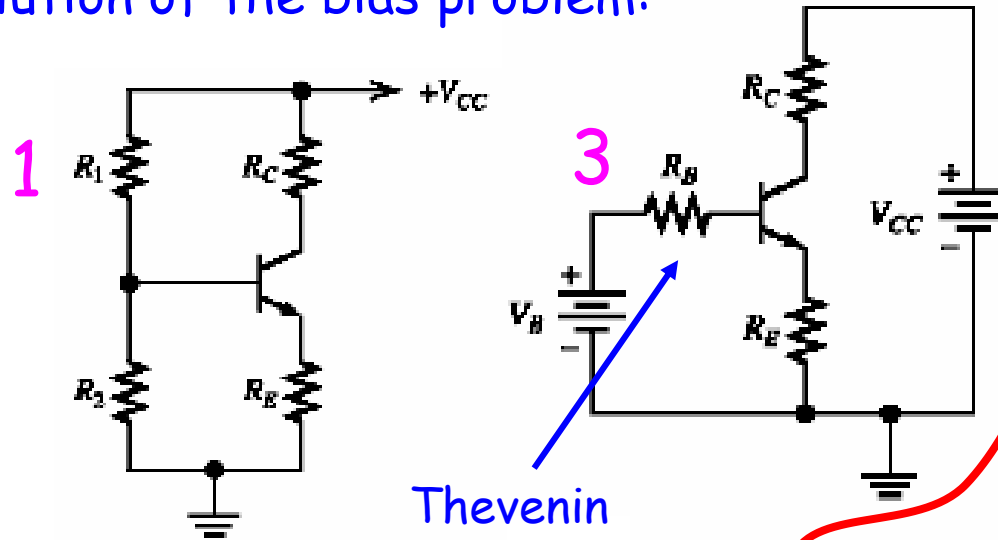
Remember: that the Q point should be independent of the β
(stability issue)

V_{BB} & V_{CC} provide this stability, however this impractical solution
Other approach is necessary to solve this problem-resistor network

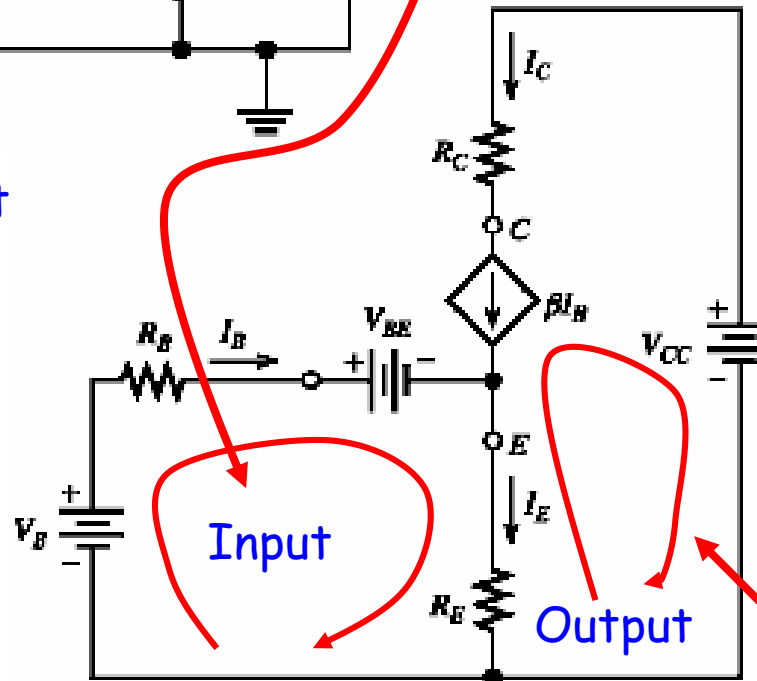
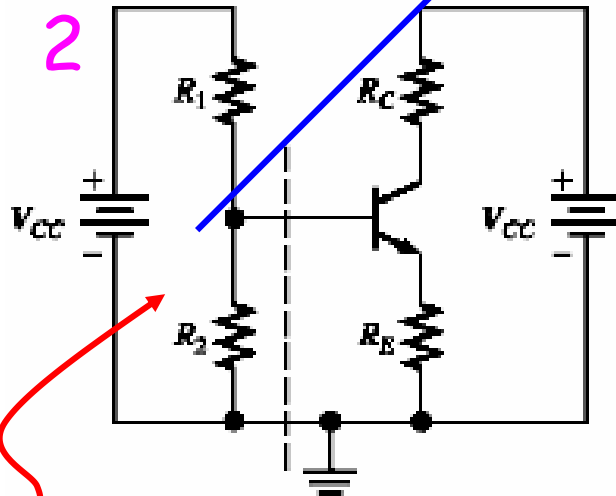


Large-Signal dc Analysis: Four-Resistor Bias Circuit

Solution of the bias problem:



Thevenin equivalent



$$V_B = R_B I_B + V_{BE} + R_E I_E$$

$$I_E = (\beta + 1) I_B \quad V_{BE} \cong 0.7V$$

$$I_B = \frac{V_B - V_{BE}}{R_B + (\beta + 1) R_E}$$

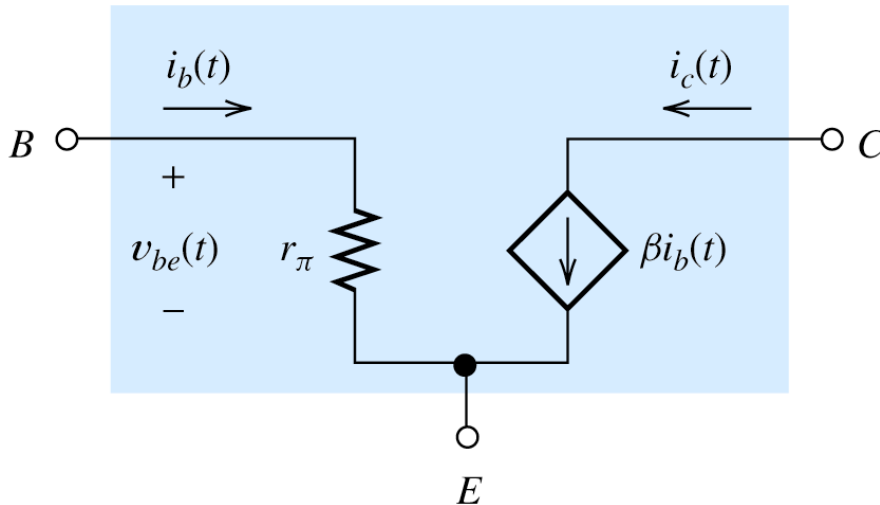
$$R_B = R_1 \parallel R_2 \quad V_B = V_{CC} \left(R_2 / (R_1 + R_2) \right)$$

$$V_{CE} = V_{CC} - R_C I_C - R_E I_E$$



Small-Signal Equivalent Circuit

Small signal equivalent circuit for BJT:



$$i_B = I_{BQ} + i_b(t) =$$

$$= (1 - \alpha) I_{ES} \left[\exp \left(\frac{v_{BEQ} + v_{be}(t)}{V_T} \right) \right]$$

$$= I_{BQ} \left[\exp \left(\frac{v_{be}(t)}{V_T} \right) \right]$$

$$\exp(x) = 1 + x,$$

$$I_{BQ} + i_b(t) = I_{BQ} \left(1 + \frac{v_{be}(t)}{V_T} \right)$$

SO

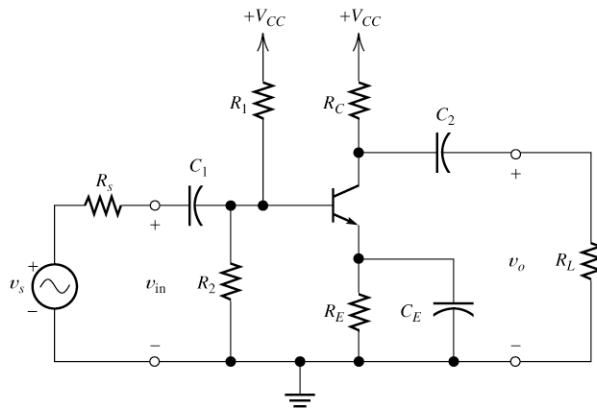
$$i_b(t) = I_{BQ} \frac{v_{be}(t)}{V_T} = \frac{v_{be}(t)}{r_\pi}$$

and

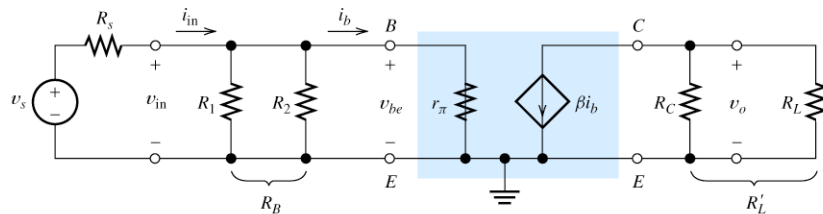
$$r_\pi = \frac{V_T}{I_{BQ}}$$



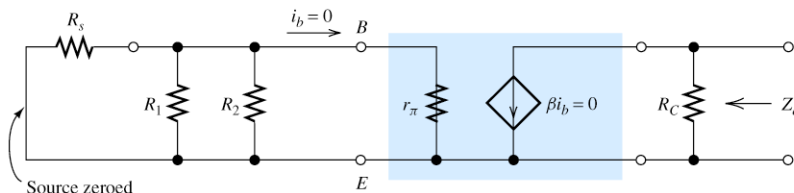
Common Emitter Amplifier



(a) Actual circuit



(b) Small-signal ac equivalent circuit



(c) Equivalent circuit used to find Z_o

First perform DC analysis to find small-signal equivalent parameters at the operating point.

Find voltage gain:

$$v_{in} = v_{be} = r_{\pi} i_b$$

$$v_o = -R'_L \beta i_b$$

$$A_v = \frac{v_o}{v_{in}} = -\frac{R'_L \beta}{r_{\pi}}$$

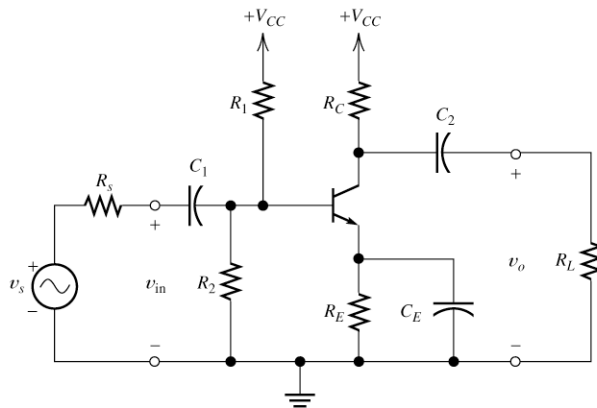
$$A_{voc} = \frac{v_o}{v_{in}} = -\frac{R_C \beta}{r_{\pi}}$$

Find input impedance:

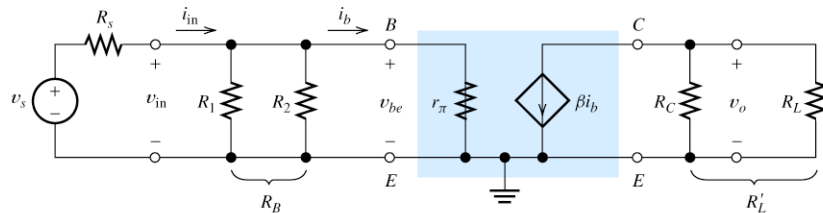
$$Z_{in} = \frac{v_{in}}{i_{in}} = \frac{1}{\frac{1}{R_B} + \frac{1}{r_{\pi}}}$$



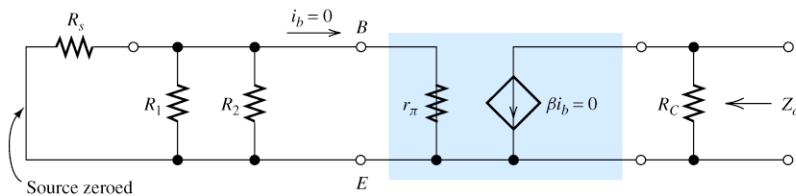
Common Emitter Amplifier



(a) Actual circuit



(b) Small-signal ac equivalent circuit



(c) Equivalent circuit used to find Z_o

Find current gain

$$A_i = \frac{i_o}{i_{in}} = \frac{A_v Z_{in}}{R_L}$$

Find power gain:

$$G = A_i A_v$$

Find output impedance:

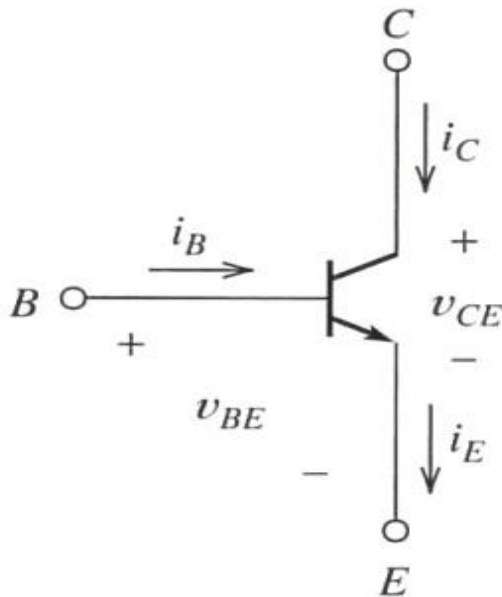
$$Z_o = R_C$$



Problem 6.6:

Suppose that a certain *npn* transistor has $V_{BE} = 0.7V$ for $I_E = 10mA$. Compute V_{BE} for $I_E = 1mA$.

Repeat for $I_E = 1\mu A$. Assume that $V_T = 26mV$.



$$I_E = I_{ES} \left(\exp\left(\frac{V_{BE}}{V_T}\right) - 1 \right) \approx I_{ES} \exp\left(\frac{V_{BE}}{V_T}\right)$$

$$10mA = I_{ES} \exp\left(\frac{0.7}{0.026}\right) \quad \text{and} \quad 1mA = I_{ES} \exp\left(\frac{V_{BE}}{0.026}\right)$$

$$\text{divide both sides} \Rightarrow 10 = \exp\left(\frac{0.7 - V_{BE}}{0.026}\right)$$

$$\Rightarrow 0.026 * \ln 10 = 0.7 - V_{BE}$$

$$\therefore V_{BE} = 0.7 - 0.026 * \ln 10 = 0.64[V]$$

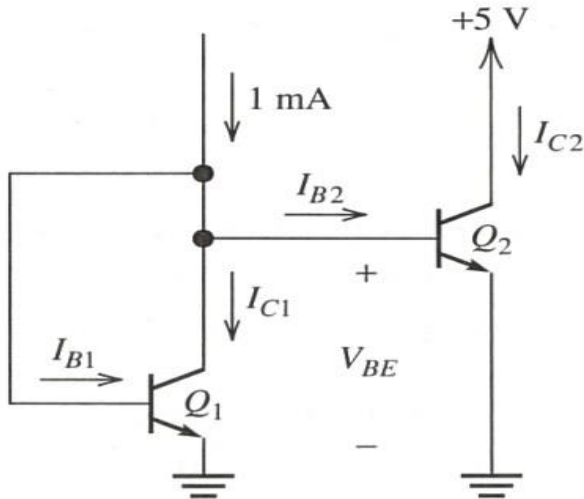


Problem 6.14:

Consider the circuit shown in Figure P6.14. Transistors Q_1 and Q_2 are identical, both having $I_{ES} = 10^{-14}A$ and $\beta = 100$. Calculate V_{BE} and I_{C2} . Assume that $V_T = 26mV$ for both transistors.

Hint: Both transistors are operating in the active region.

Because the transistors are identical and have identical values of V_{BE} , their collector currents are equal.



$$I_{B1} + I_{B2} + I_C = 1mA \quad \& \quad I_C = \beta I_B$$
$$\Rightarrow I_C \left(\frac{2}{\beta} + 1 \right) = 1mA \Rightarrow I_C = \frac{1mA}{1.02} = 0.98mA$$

$$I_E = \left(1 + \frac{1}{\beta} \right) I_C = 0.99mA$$

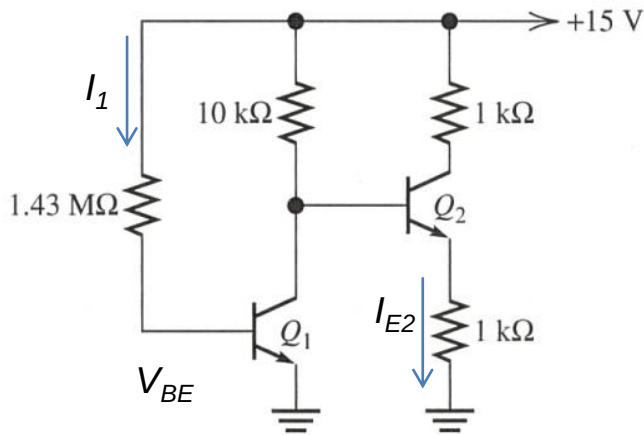
$$\text{since } I_E \approx I_{ES} \exp\left(\frac{V_{BE}}{V_T}\right) \text{ we have}$$

$$\therefore V_{BE} = V_T \ln \frac{I_E}{I_{ES}} = 0.026 * \ln(0.99 * 10^{11}) = 0.658V$$



Problem 6.50:

The transistors shown in Figure P6.50 operate in active region and have $\beta = 100$, $V_{BE} = 0.7V$. Determine I_C and V_{CE} for each transistor.



$$I_1 = \frac{14.3}{1.43M\Omega} = 10\mu A$$

$$I_{C1} = \beta I_1 = 1mA$$

$$\frac{(15 - (I_{E2} * 1k + 0.7))}{10k} = I_{C1} + \frac{I_{E2}}{\beta + 1}$$

$$\frac{14.3}{10k} - \frac{I_{E2}}{10} = 1mA + \frac{I_{E2}}{101}$$

$$0.43mA = I_{E2} * \left(\frac{1}{101} + \frac{1}{10} \right)$$

$$I_{E2} = 3.9126mA \Rightarrow I_{C2} = 0.99I_{E2} = 3.8735mA$$

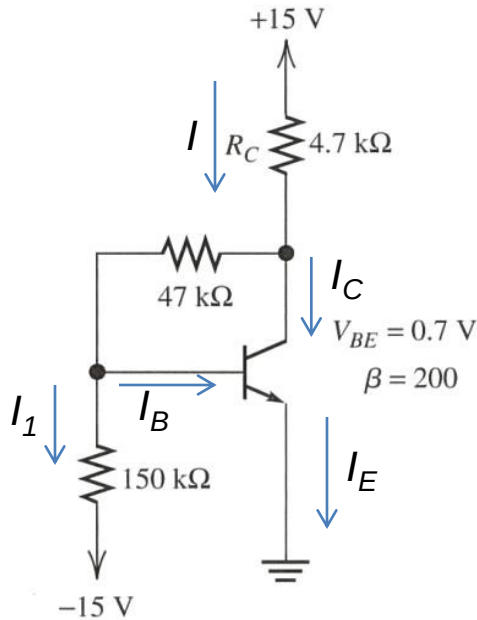
$$V_{CE2} = 15 - 1k * (I_{C2} + I_{E2}) = 15 - 1.99k * I_{E2} = 7.213V$$

$$V_{CE1} = 15 - 10k * \left(1mA + \frac{I_{C2}}{\beta} \right) = 4.6126V$$



Problem 6.52:

Analyze the circuit of Figure P6.52 to determine I_C and V_{CE} .



$$I = I_1 + I_E$$

$$I_C = \beta I_B$$

$$I_1 = \frac{(0.7 - (-15))V}{150K\Omega} = \frac{(0.7 + 15)V}{150K\Omega} = 0.1047mA$$

$$I_E = (\beta + 1)I_B$$

$$(I_1 + I_B) * 47k\Omega + 0.7 + I * 4.7k\Omega = 15V$$

$$I_1 * 47k + I_B * 47k + 0.7 + I_1 * 4.7k + (\beta + 1) * I_B * 4.7k = 15V$$

$$I_1 * (47k + 4.7k) = 15 - 0.7 - I_B * (47k + 201 * 4.7k)$$

$$I_B = \frac{14.3 - 0.1047mA * 51.7k}{47k + 944.7k} = \frac{14.3 - 5.413}{991.7k} = 9.0\mu A$$

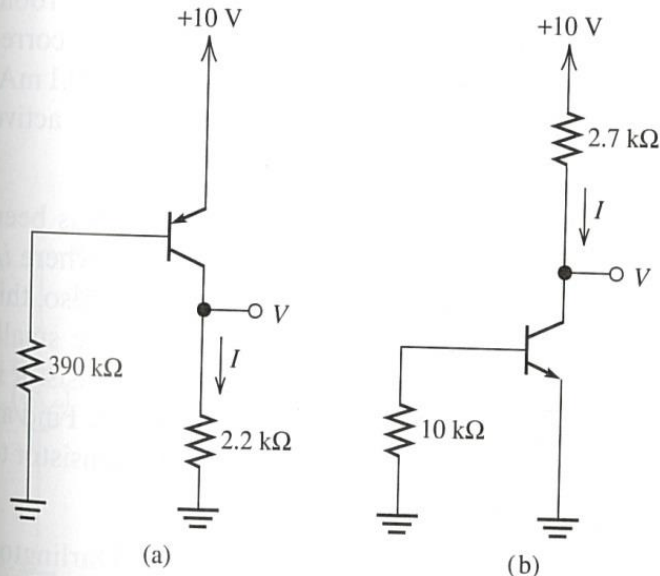
$$I_C = \beta I_B = 1.8mA$$

$$V_{CE} = V_{BE} + 47k * (I_1 + I_B) = 0.7 + 47k * 0.1137mA = 6.04V$$



Problem 6.45:

Analyze the circuits shown in Figure P6.45 to determine I and V . For all transistors, assume that $\beta = 100$ and $|V_{BE}| = 0.7V$ in both the active and saturation regions. Repeat for $\beta = 300$.



(a) for $\beta = 100$

$$|V_{BE}| = 0.7 \Rightarrow V_B = 9.3V$$

$$I_B = \frac{9.3}{390k} = 23.8\mu A$$

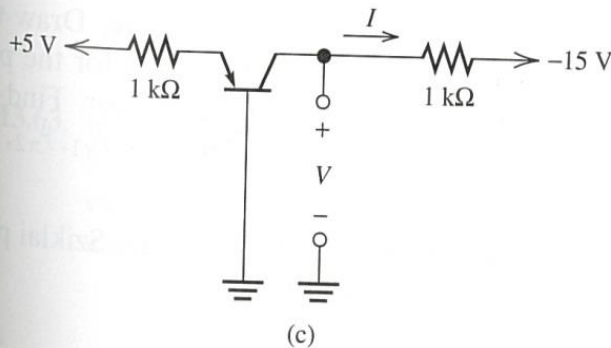
$$I_C = \beta I_B = 2.38mA \Rightarrow V = I_C * 2.2k = 5.236V$$

for $\beta = 300$

$$I_C = 7.15mA \Rightarrow V = I_C * 2.2k = 15.73V \quad (\text{Incorrect})$$

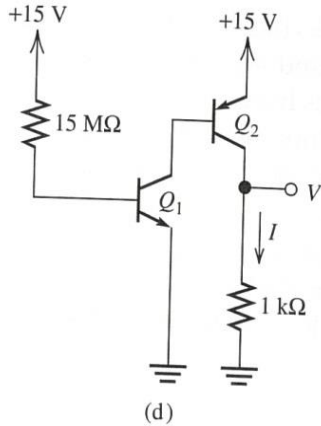
$$\text{Since } V_{\max} = 9.8V \Rightarrow I_{C\max} = \frac{9.8}{2.2k} = 4.43mA$$

$$\Rightarrow \beta_{\max} = \frac{I_{\max}}{I_B} = \frac{4.42mA}{23.8\mu A} = 187.2$$



Problem 6.45: Contd.

(d) For $\beta = 100$



$$I_{B1} = \frac{14.3}{15M\Omega} = 0.9533\mu A$$

$$I_{C1} = \beta I_{B1} = 95.33\mu A = I_{B2}$$

$$I = I_{C2} = I_{B2} * \beta = 9.533mA \quad \Rightarrow \quad V = I * 1k = 9.533V$$

For $\beta = 300$, $I_{B2} = 286\mu A \quad \Rightarrow \quad I_{C2} = I_{B2} * \beta = 85.8mA$
(would give $V = 85.8V$, Incorrect)

$$I_{C2} = \beta^2 I_{B1} \quad \text{and since } V_{\max} = 14.8V$$

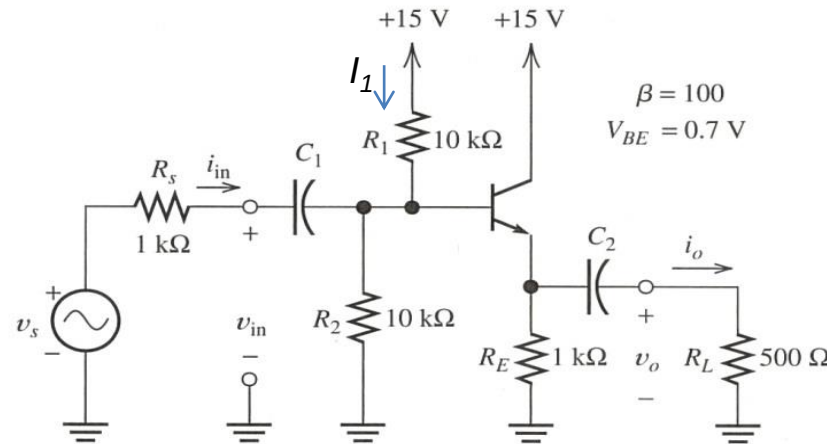
$$I_{\max} = 14.8mA = I_{C2\max}$$

$$\beta_{\max} = \sqrt{\frac{I_{C2\max}}{I_{B1}}} = \sqrt{\frac{14.8mA}{0.953\mu A}} = 124.5$$



Problem 6.67:

Consider the emitter-follower amplifier of Figure P6.67 . Draw the dc circuit and find I_{CQ} . Next, determine the value of r_{π} . Then, calculate midband values for A_v , A_{voc} , Z_{in} , A_i , G and Z_o .



DC Analysis

$$I_1 * 10k + (I_1 - I_{BE}) * 10k = 15V \quad \Rightarrow \quad I_1 * 20k - I_B * 10k = 15V$$

$$15 - I_1 * 10k - 0.7 = (1 + \beta) * I_B * 1k\Omega \quad \Rightarrow \quad I_1 * 10k + I_B * 101k = 14.3V$$

multiply 2nd equation by 2 and subtract the first one

$$I_B * (202k + 10k) = 28.6 - 15$$

$$I_B = \frac{13.6}{212k} = 64.2\mu A$$

$$I_{CQ} = I_B * \beta = 6.42mA$$



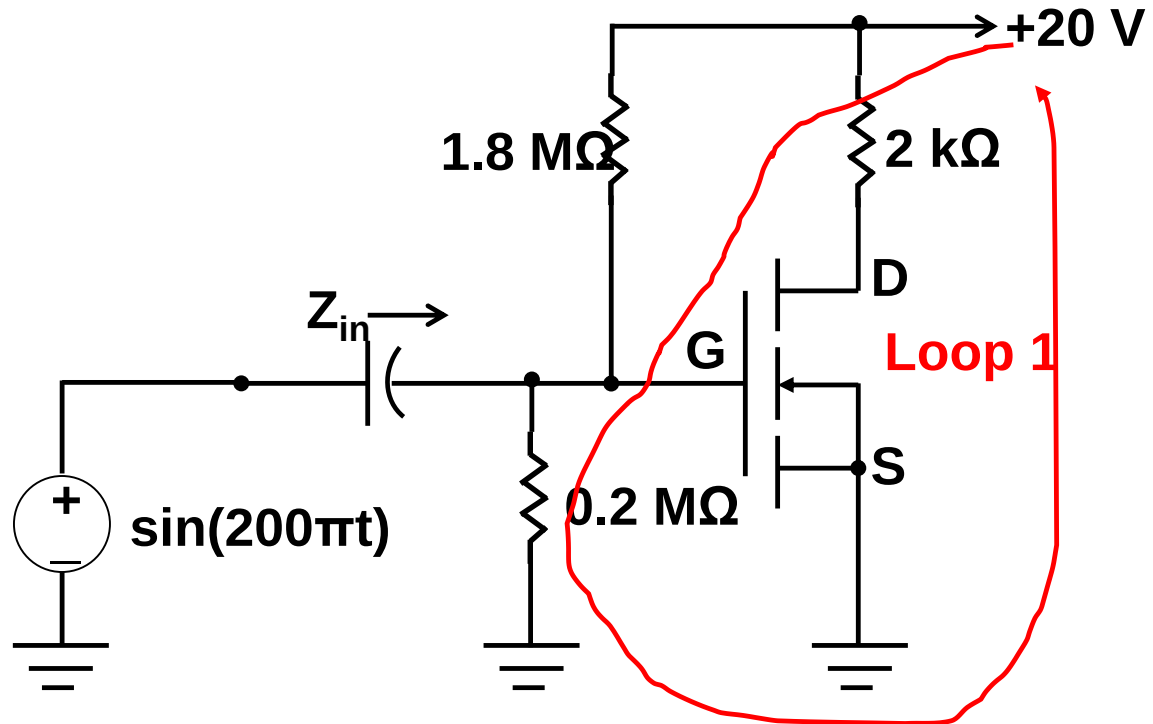
Problem 8: Consider the amplifier shown below.

a) Find $v_{GS}(t)$. Assume that the coupling capacitor is a short circuit for the ac signal and an open circuit for the dc.

Soln (a): In loop 1 the 1.8 M Ω and 200 k Ω resistors act as voltage divider.

The voltage drop across 200 k Ω resistor is the dc voltage V_{GSQ}

$$V_{GSQ} = 20 \times 0.2 / 2 = 2 \text{ V}$$



Treating the capacitor as short for ac signals, we have

$$V_{GS} = 2 + \sin(200\pi t)$$

- b) If the FET has $V_{t0} = 1V$ and $K = 0.5 \text{ mA/V}^2$, sketch its drain characteristics to scale for $V_{GS} = 1, 2, 3$, and 4 V .
- c) Draw the load line for the amplifier on the characteristics.
- d) Find the values of V_{DSQ} , V_{DSmin} , and V_{DSmax} .

To obtain the drain characteristics apply the following equations

$$i_D = \begin{cases} 0 & \text{when } v_{GS} < V_{t0} \\ K[2(v_{GS} - V_{t0})v_{DS} - v_{DS}^2] & \text{when } (V_{GS} - V_{t0}) > 0 \\ K(v_{GS} - V_{t0})^2 & \text{when } v_{DS} > (v_{GS} - V_{t0}) > 0 \end{cases}$$



$$i_D = \begin{cases} 0 & \text{when } v_{GS} < V_{t0} \\ K[2(v_{GS} - V_{t0})v_{DS} - v_{DS}^2] & \text{when } (V_{GS} - V_{t0}) > 0 \\ K(v_{GS} - V_{t0})^2 & \text{when } v_{DS} > (v_{GS} - V_{t0}) > 0 \end{cases}$$

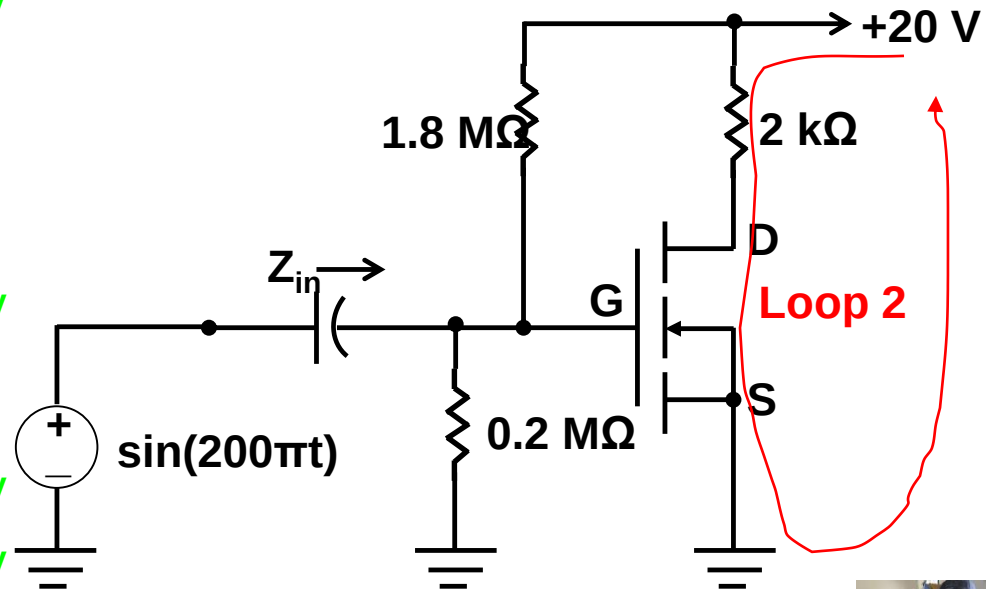
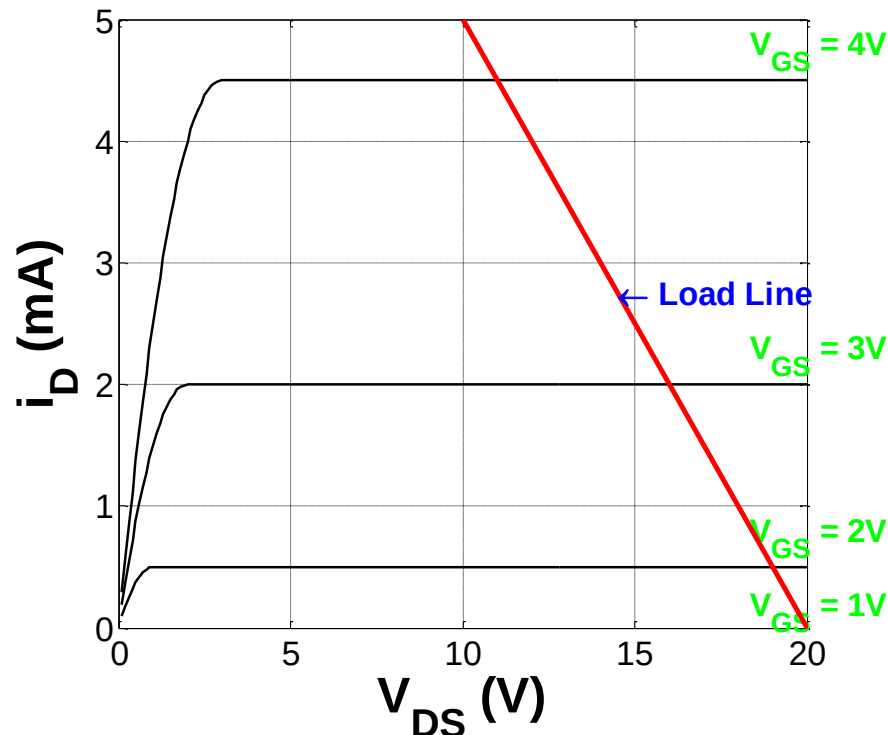
b) Plot shows the drain characteristics for $V_{GS} = 1, 2, 3$, and 4 V.

c) To get the load line apply KVL to loop 2:

$$20 - 2 \text{ k}\Omega \cdot i_D(t) = V_{DS}(t)$$

The red line in the plot is the load line.

Drain Characteristics



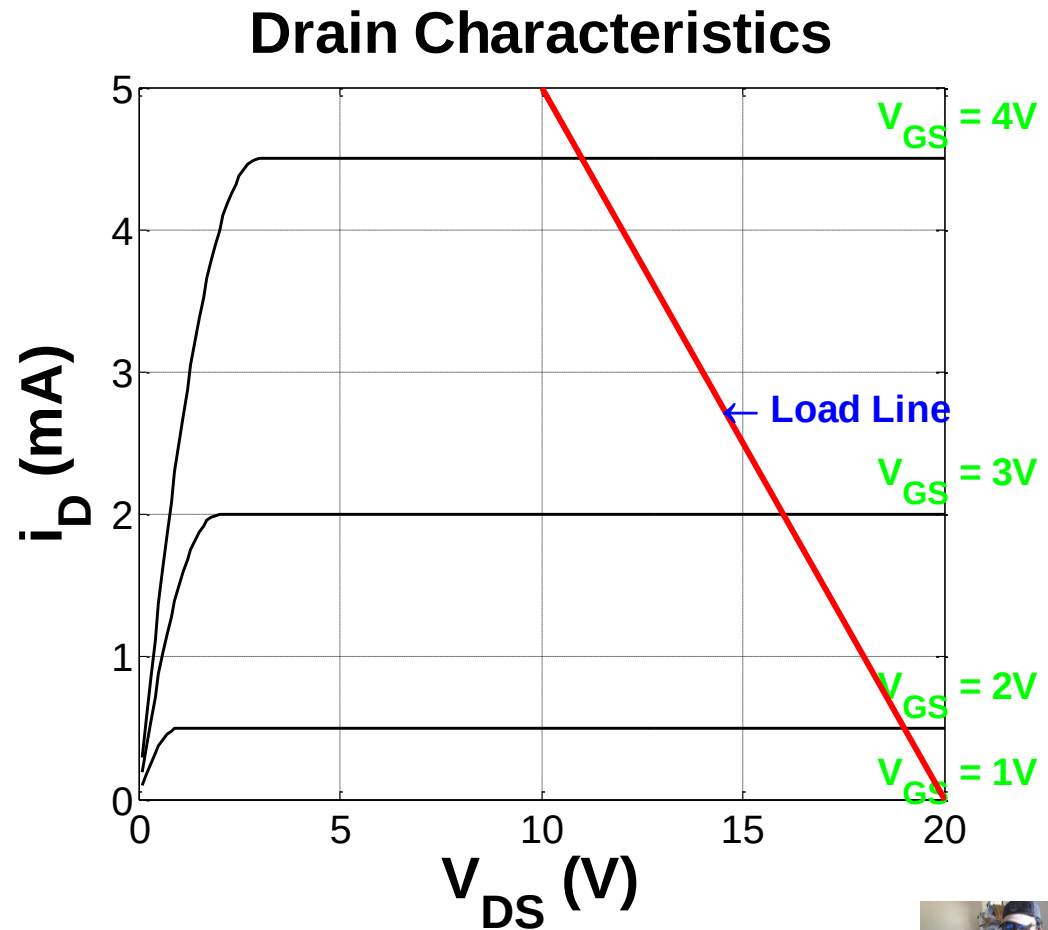
d) Find the values of V_{DSQ} , V_{DSmin} , and V_{DSmax} .

d) V_{DSQ} , V_{DSmin} , and V_{DSmax} are the points at which the load line intersects the drain characteristics for $V_{GS} = 2\text{ V}$, 3 V and 1 V respectively.

$$V_{DSQ} = 19\text{ V}$$

$$V_{DSmin} = 16\text{ V}$$

$$V_{DSmax} = 20\text{ V}$$



Problem 9: Consider the common source amplifier shown below. Assume NMOS transistor has the following parameters:

$KP=60 \mu A/V^2$, $L=5 \mu m$, $W=100 \mu m$, $r_d=\infty$, and $V_{to}=1.5 V$.

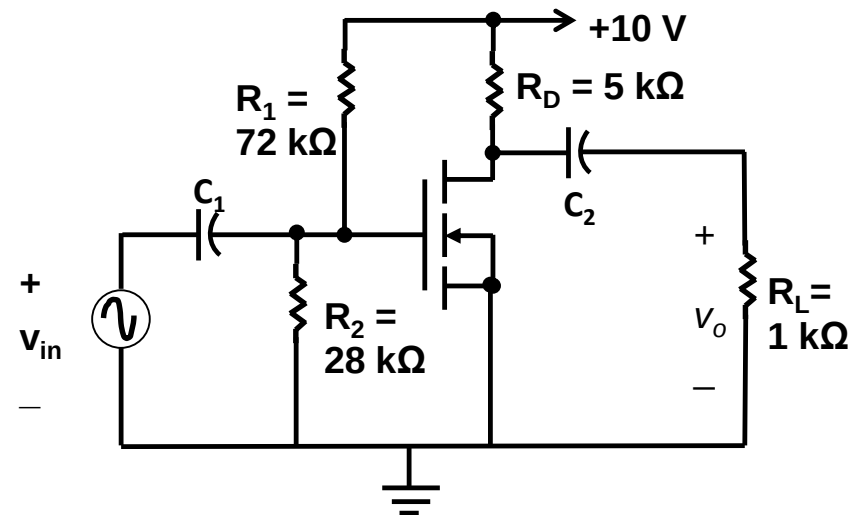
a) Find the values of I_{DQ} , V_{DSQ} and g_m

The $72 k\Omega$ and $28 k\Omega$ resistors act as a voltage divider. The voltage drop across $28 k\Omega$ resistor is the dc voltage V_{GSQ} is equal to

$$V_{GSG} = V_{DD} \frac{R_2}{R_1 + R_2} = 10 \frac{28}{72 + 28} = 2.8V$$

$$K = \frac{1}{2} KP \left(\frac{W}{L} \right) = 0.6 mA/V^2$$

$$I_{DQ} = K (V_{GSQ} - V_{to})^2 = 1.014 mA$$



$$V_{DSQ} = (V_{DD} - R_D I_{DQ}) = 4.93V$$

$$g_m = 2\sqrt{KI_{DQ}} = 1.56 mS$$



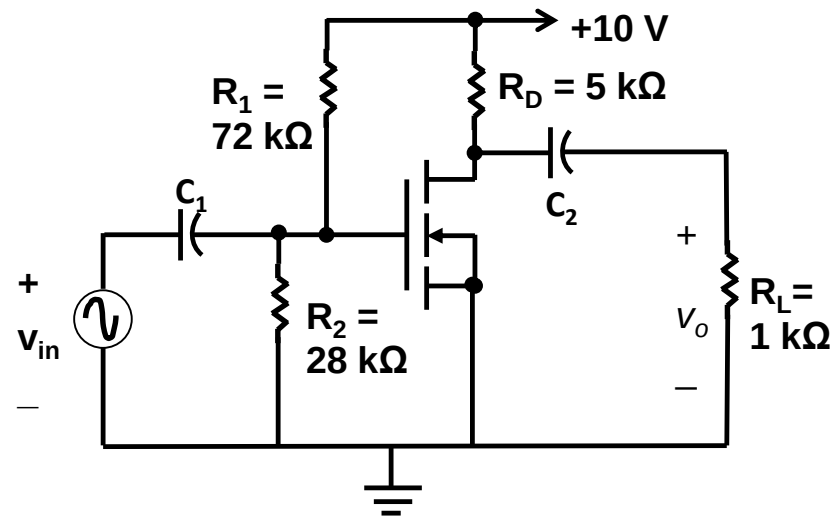
Problem 9 b): - Assuming that the coupling capacitors are short circuits for the ac signal, determine the following: voltage gain, input resistance and output resistance.

$$R_L' = \frac{1}{\frac{1}{R_D} + \frac{1}{R_L}} = 833.3\Omega$$

$$A_v = -g_m R_L' = -1.3$$

$$R_{in} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = 20.16k\Omega$$

$$R_o = R_D = 5k\Omega$$

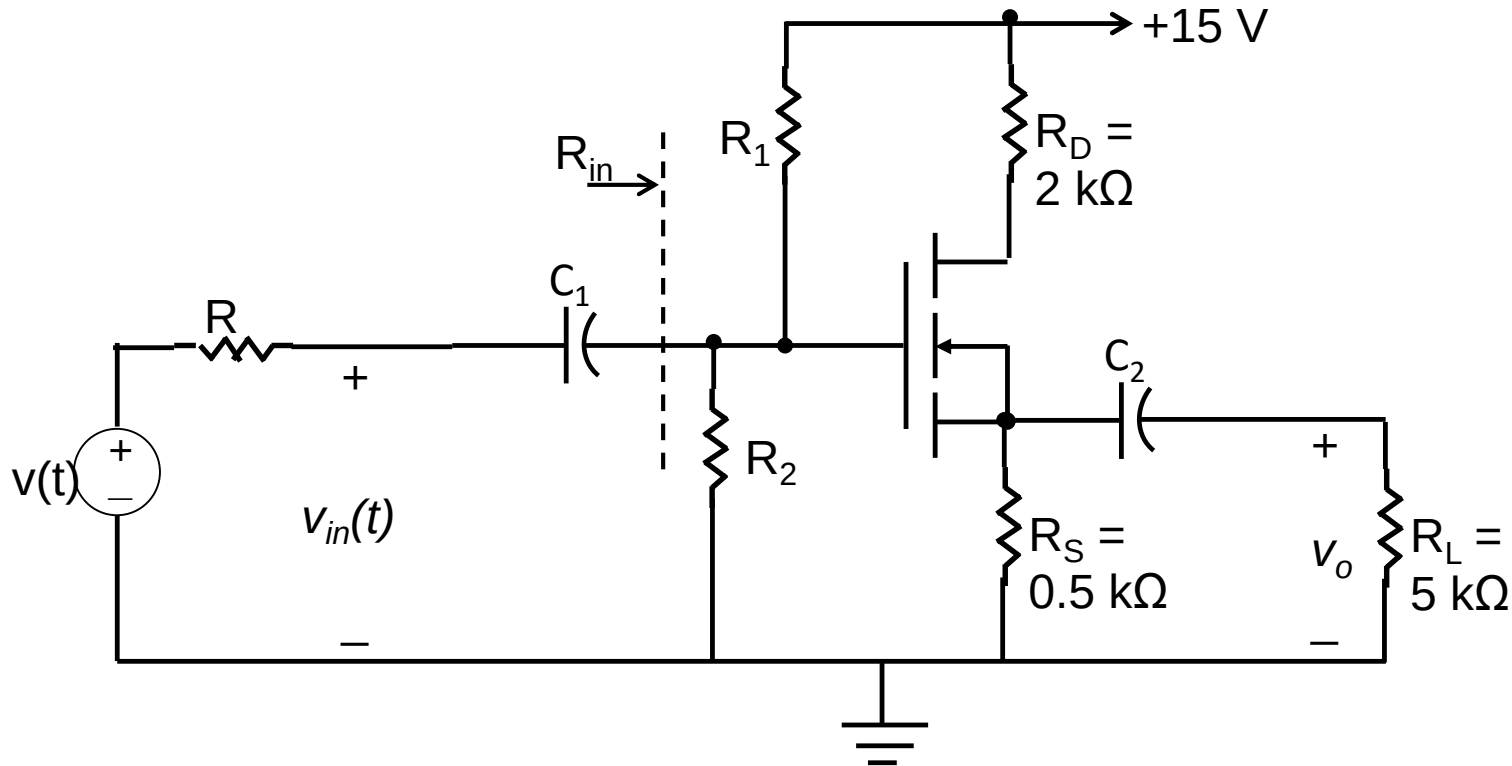


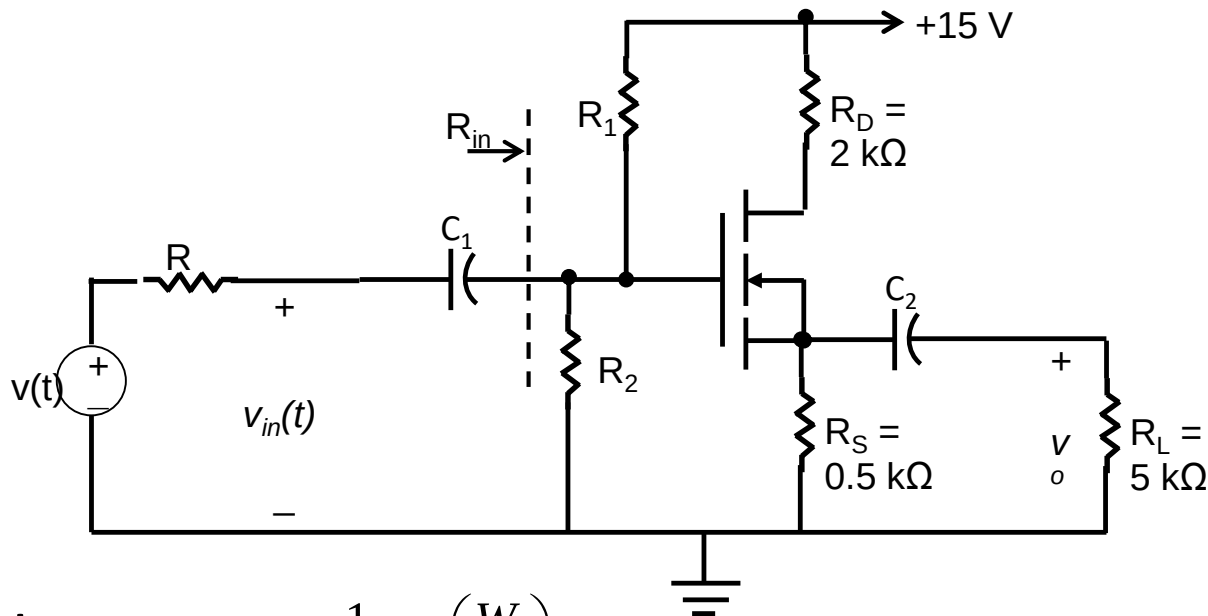
Problem 10: - Consider the common source amplifier shown below.

Assume NMOS transistor has the following parameters:

$KP=75 \mu A/V^2$, $L=10 \mu m$, $W=400 \mu m$, $r_d=\infty$, and $V_{to}=1 V$.

a) If $R_{in} = 250 k\Omega$, find the values for R_1 and R_2 to achieve $I_{DQ}=2 mA$.





- We have: $K = \frac{1}{2} KP \left(\frac{W}{L} \right) = 1.5 \text{ mA/V}^2$
- Given:

$$I_{DQ} = K(V_{GSQ} - V_{to})^2 = 2 \text{ mA} \quad \longrightarrow \quad V_{GSQ} = V_{to} + \sqrt{I_{DQ}/K} = 2.155 \text{ V}$$

$$V_S = R_S I_{DQ} = 1 \text{ V} \quad V_G = V_{GSQ} + V_S = 3.155 \text{ V}$$

$$V_G = V_{DD} \frac{R_2}{R_1 + R_2} = V_{DD} \frac{1}{R_1} R_{in}$$

- Solve for R_1 : $R_1 = V_{DD} \frac{1}{V_G} R_{in} = 15 * \frac{1}{3.155} * 250 * 10^3 = 1.19 \text{ M}\Omega$



- We have $R_{in} = 250 \text{ k}\Omega$ and $R_1 = 1.19 \text{ M}\Omega$

$$R_{in} = (R_1 \parallel R_2) = \frac{R_1 * R_2}{R_1 + R_2}$$

- Solve for R_2 : $250k = \frac{1.19M * R_2}{1.19M + R_2} \Rightarrow R_2 = 316.5k\Omega$

b) Determine the voltage gain

$$R_L' = \frac{1}{\frac{1}{R_d} + \frac{1}{R_L} + \frac{1}{R_S}} = 454.54\Omega$$

$$g_m = 2\sqrt{KI_{DQ}} = 3.46\text{mS}$$

$$A_v = \frac{V_0}{V_{in}} = -g_m R_L' = -1.572$$



Problem BJT P1: It has been found that in the circuit below $V_E = 1V$. If $V_{BE} = -0.6V$, determine: V_B , I_B , I_E , I_C , β , and α .

Soln (a): From KVL: $5V = I_E * R_E + 1V$

$$\Rightarrow 4V = I_E * 5000 \Rightarrow I_E = 0.8mA$$

From KVL: $V_B = V_E - V_{EB} = 1 - 0.6 = 0.4V$

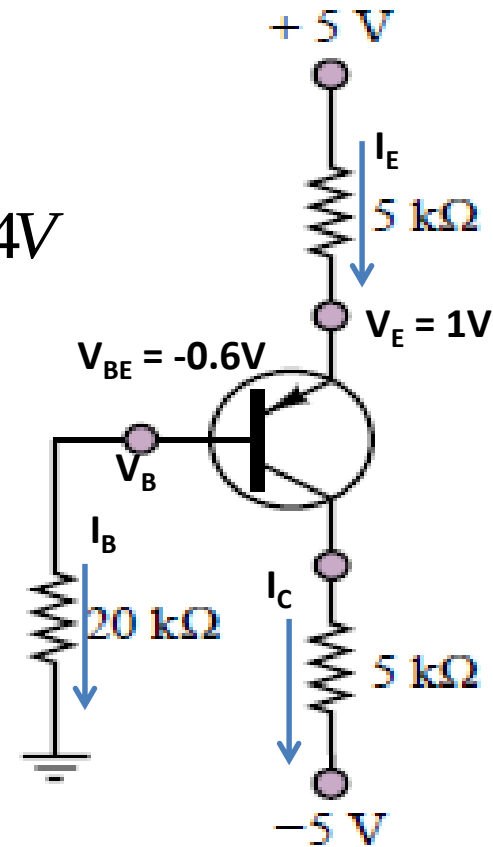
Ohm's law: $V_B = I_B * R_B$

$$0.4V = I_B * 20k\Omega \Rightarrow I_B = 20\mu A$$

$$I_C = I_E - I_B = 0.78mA$$

$$\beta = \frac{I_C}{I_B} = 39$$

$$\alpha = \frac{I_C}{I_E} = 0.975$$



Problem BJT P2: - For the circuit below assume both transistors are silicon-based with $\beta = 100$. Determine: **a)** I_{C1} , V_{C1} , V_{CE1} . **b)** I_{C2} , V_{C2} , V_{CE2} .

• **Soln:**

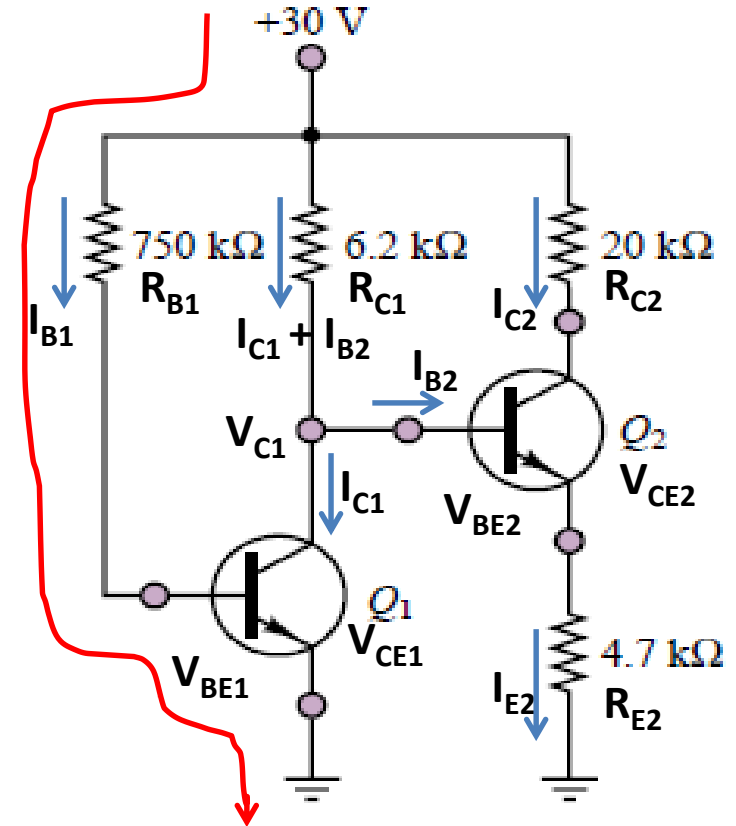
Assume $V_{BE} = V_{BE1} = V_{BE2} = 0.7V$

- **Part (a):** - Apply KVL along the path (red line).

$$-30 + I_{B1} * R_{B1} + V_{BE1} = 0$$

$$I_{B1} = \frac{30 - 0.7}{750 * 10^3} = 39.07 \mu A$$

$$I_{C1} = \beta * I_{B1} = 3.907 mA$$



- Part (a) contd.:** - Apply KVL along the path (red line).

$$30 - (I_{C1} + I_{B2})R_{C1} - V_{BE2} - I_{E2}R_{E2} = 0$$

We know that $I_E = (\beta + 1)I_B$

substituting we get

$$30 - 24.2234 - I_{B2}R_{C1} - 0.7 - (\beta + 1)I_{B2}R_{E2} = 0$$

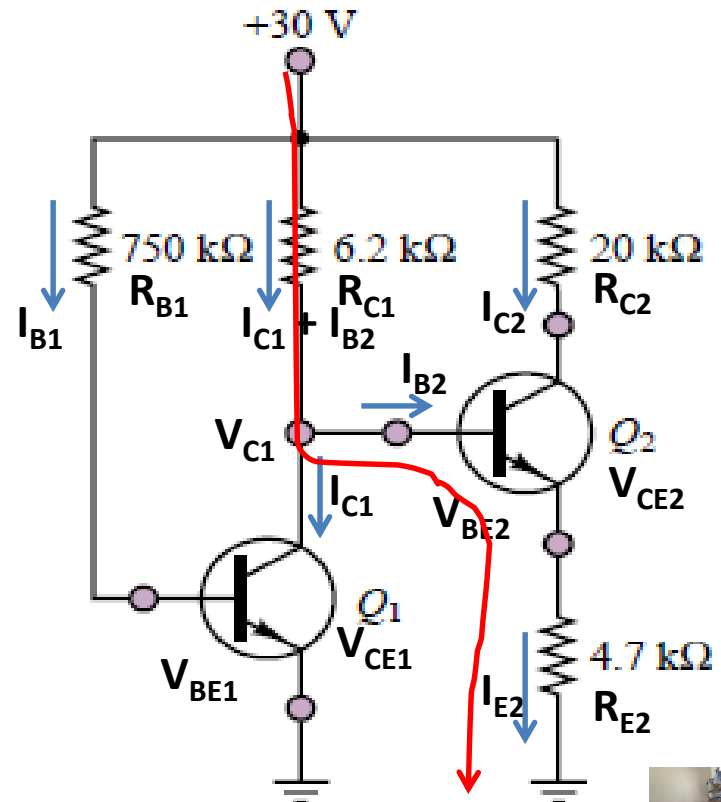
$$5.0766 - I_{B2}(R_{C1} + 101 * R_{E2}) = 0 \Rightarrow I_{B2} = 10.559 \mu A$$

$$V_{C1} = 30 - (I_{C1} + I_{B2}) * R_{C1}$$

$$V_{C1} = 30 - (3.907 + 0.010559) * 6.2$$

$$= 5.7111 [V]$$

$$V_{CE1} = V_{C1} = 5.7111V$$



- $$I_{E2} = (1 + \beta)I_{B2} = 101 * 10.559 \mu A = 1.0662 mA$$

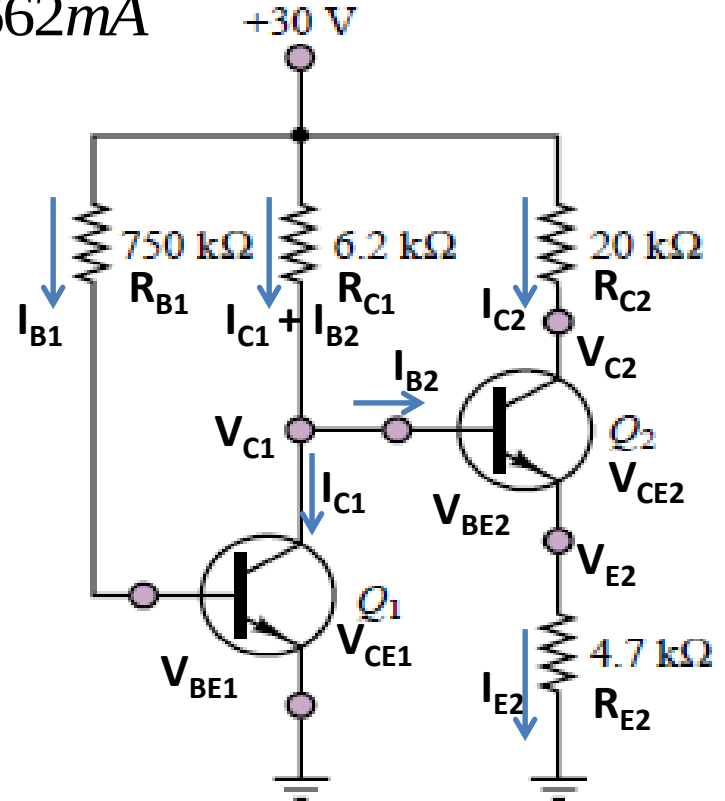
$$I_{C2} = I_{B2}\beta = 1.0556mA$$

$$V_{C2} = 30 - I_{C2}R_{C2}$$

$$V_{C_2} = 30 - 1.0556 * 20 = 8.888V$$

$$V_{E2} = I_{E2} R_{E2} = 5.0111V$$

$$V_{CE2} = V_{C2} - V_{E2} = 3.8769V$$



Problem BJT P3: - Design the bias circuit (find R_C and R_B) to give a Q-point of $I_C = 20\mu\text{A}$ and $V_{CE} = 0.9\text{V}$ if the transistor current gain $\beta_F = 50$ and $V_{BE} = 0.65\text{V}$.

What is the Q-point if the current gain of the transistor is 125?

- **Soln:** Apply KVL along the path (red line).

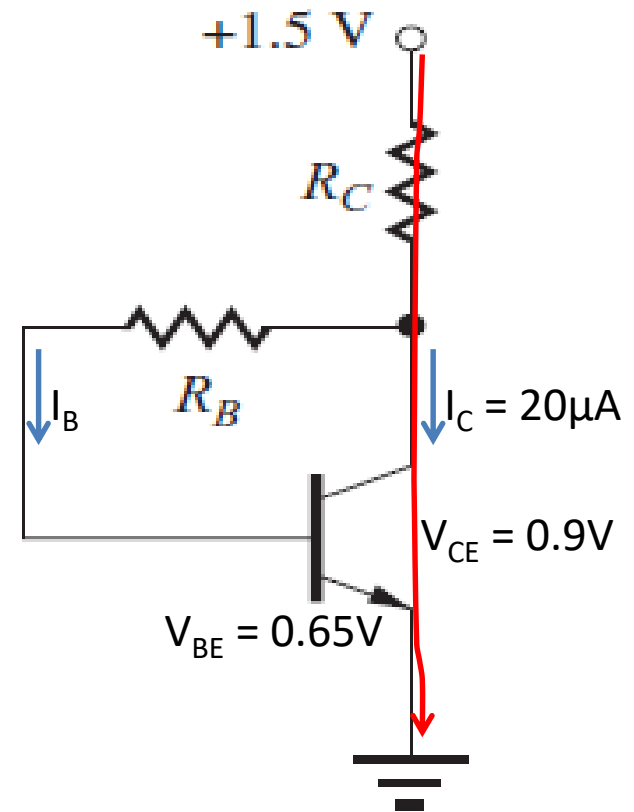
$$(I_C + I_B)R_C + V_{CE} = 1.5$$

$$\left(I_C + \frac{I_C}{\beta}\right)R_C + 0.9 = 1.5$$

$$I_C \left(1 + \frac{1}{\beta}\right)R_C = 0.6$$

$$20 * 10^{-6} \left(1 + \frac{1}{50}\right)R_C = 0.6$$

$$R_C = \frac{0.6}{20.4 * 10^{-6}} = 29.4117\text{k}\Omega$$



- **Soln contd.:** (find R_C and R_B) to give a Q-point of $I_C = 20\mu\text{A}$ and $V_{CE} = 0.9\text{V}$.
- Apply KVL along the path (red line).

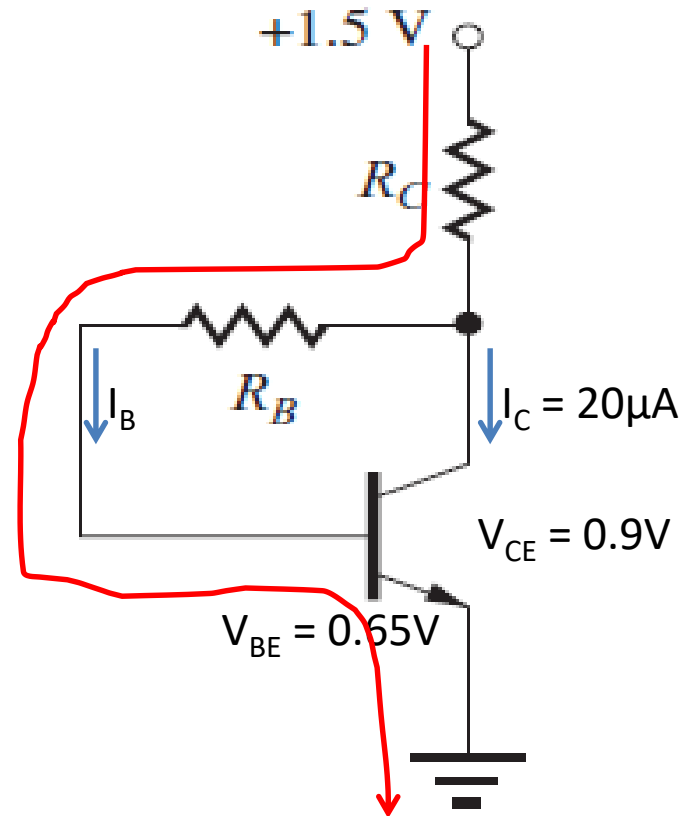
$$(I_C + I_B)R_C + I_B R_B + V_{BE} = 1.5$$

$$\left(I_C + \frac{I_C}{\beta}\right)R_C + \frac{I_C}{\beta}R_B + 0.65 = 1.5$$

$$0.6 + \left(\frac{20 * 10^{-6}}{50}\right)R_B + 0.65 = 1.5$$

$$0.4 * 10^{-6} * R_B = 0.25$$

$$R_B = \frac{0.25}{0.4 * 10^{-6}} = 625k\Omega$$



- **Soln contd.:** Find the Q-point if the current gain, $\beta_F = 125$. We have $R_C = 29.41\text{k}\Omega$, and $R_B = 625\text{k}\Omega$, from previous calculations.
- Apply KVL along the path (red line).

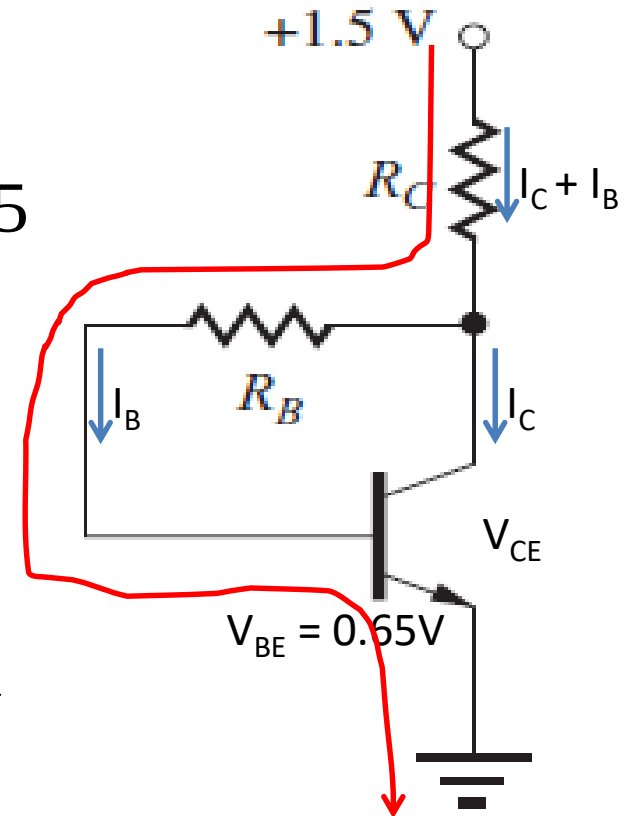
$$(I_C + I_B)R_C + I_B R_B + V_{BE} = 1.5$$

$$(\beta I_B + I_B) 29.41\text{k} + I_B * 625\text{k} = 0.85$$

$$(126 * 29.41\text{k} + 625\text{k}) I_B = 0.85$$

$$I_B = \frac{0.85}{4.331 * 10^6} = 0.196\mu\text{A}$$

$$I_C = \beta I_B = 125 * 0.196 * 10^{-6} = 24.53\mu\text{A}$$



- **Soln contd.:** Apply KVL along the path (red line).

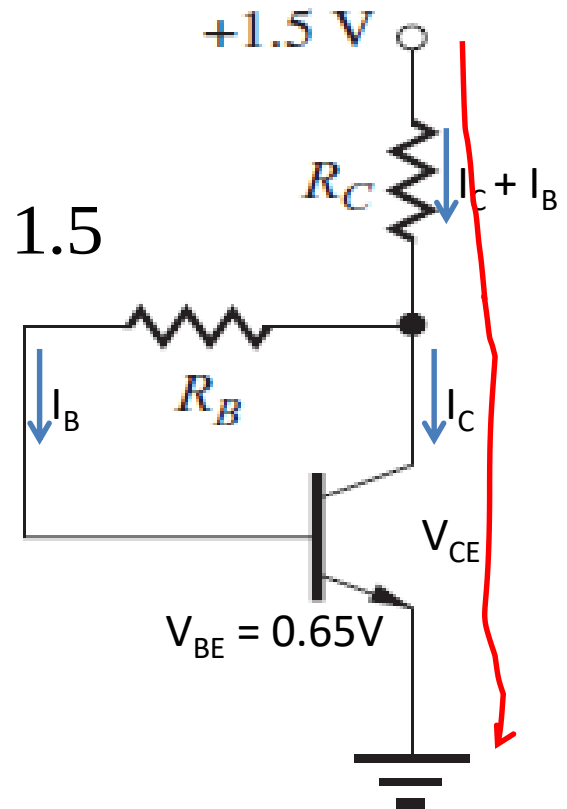
$$(I_C + I_B)R_C + V_{CE} = 1.5$$

$$(24.53 + 0.196) * 10^{-6} * 29.41 k + V_{CE} = 1.5$$

$$V_{CE} = 1.5 - 0.727 = 0.773V$$

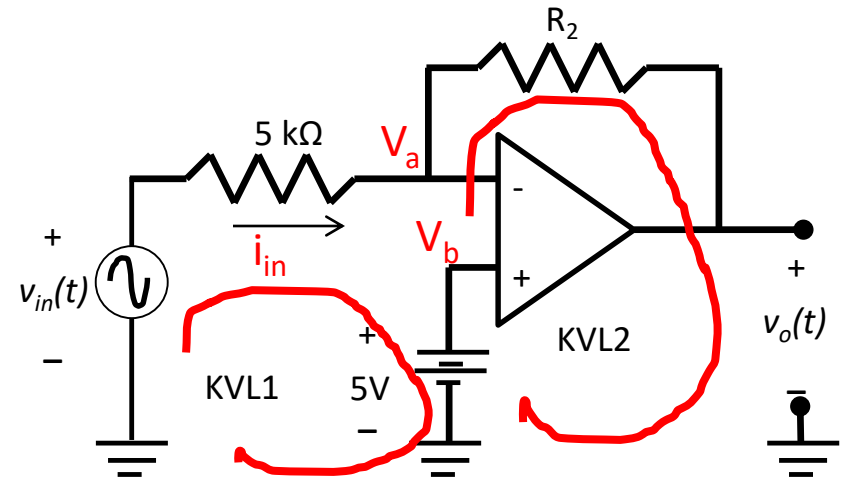
- The Q-Point is:

$$(I_C, V_{CE}) = (24.53 \mu A, 0.773V)$$



Problem OP-AMP P1: - Consider the op-amp circuit shown below. If $v_{in}(t) = 6 + 9\cos(500\pi t)$, calculate the value of R_2 required to generate a output, $v_o(t)$, with zero DC component. What is the resulting output voltage?

- **Soln:** The circuit shown is that of a differential amplifier. We can use superposition theorem to solve for the output voltage: connect inputs to ground (0 V), one at a time, and solve for output voltage.



- From summing point constraints: $V_a = V_b$
- From KVL2 $v_o = 5 - i_{in}(t)R_2$
- From KVL1 and Ohms law $i_{in} = \frac{v_{in} - 5}{5k\Omega}$
- Therefore

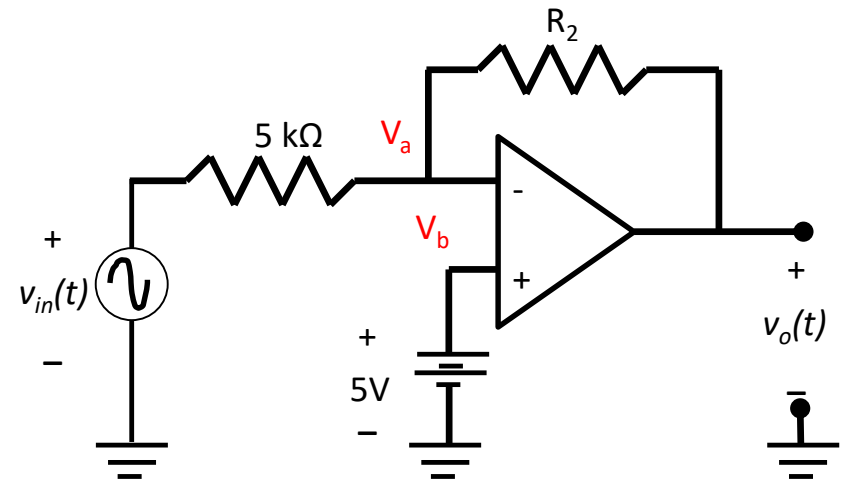
$$v_o = 5 - \frac{(6 + 9\cos(500\pi t)) - 5}{5k\Omega} * R_2$$



$$v_o = 5 - \frac{(6 + 9\cos(500\pi t)) - 5}{5k\Omega} * R_2$$

- If DC component of v_o is zero,

$$0 = 5 - \frac{6 - 5}{5k\Omega} * R_2$$



- Multiplying by $5k\Omega$ on both sides and solving for R_2 ,
 $R_2 = 25k\Omega$
- Then the output is $v_o = -45\cos(500\pi t)$,

Problem OP-AMP P2: - Consider the op-amp circuit shown below. Assume the maximum output voltage of the op-amp ranges from -12 V to $+12\text{ V}$; the maximum output current magnitude is 25 mA ; and the slew-rate limit is $1.5\text{ V}/\mu\text{s}$. If $v_{in}(t) = v_m \sin(\omega t)$, $R_1 = 5\text{ k}\Omega$, and $R_2 = 25\text{ k}\Omega$.

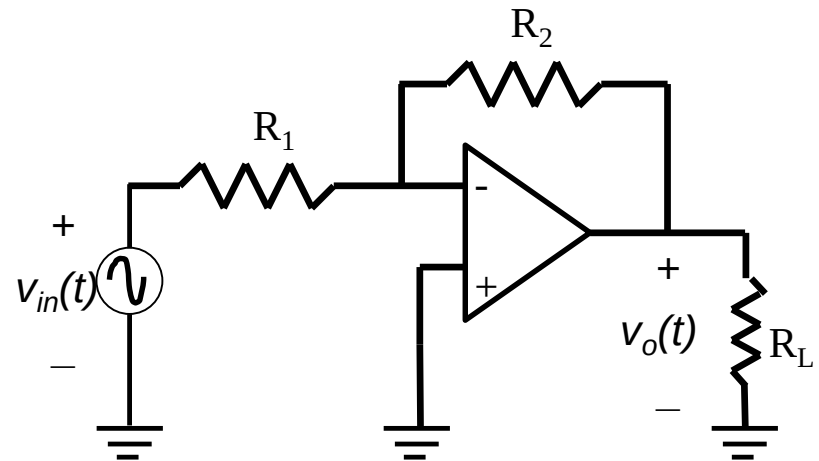
a) Find the full-power bandwidth of the op-amp.

- Soln:** The full-power bandwidth of the op-amp is given by

$$f_{FP} = \frac{SR}{2\pi V_{om}}$$

- Slew-rate, $SR = 1.5\text{ V}/\mu\text{s}$;
maximum output amplitude, $V_{om} = 12\text{ V}$.

$$f_{FP} = \frac{1.5 \times 10^6}{2\pi(12)} \approx 19.9\text{ kHz}$$



b) Find the peak output voltage possible without distortion for the following cases:

- **Case a:** Frequency of 5 kHz and $R_L = 20\ \Omega$
 - **Soln.:** The current limit of the op-amp limits the peak output voltage. Since R_L is very small compared to R_2 the current through R_2 can be neglected. Thus the peak output voltage is given by

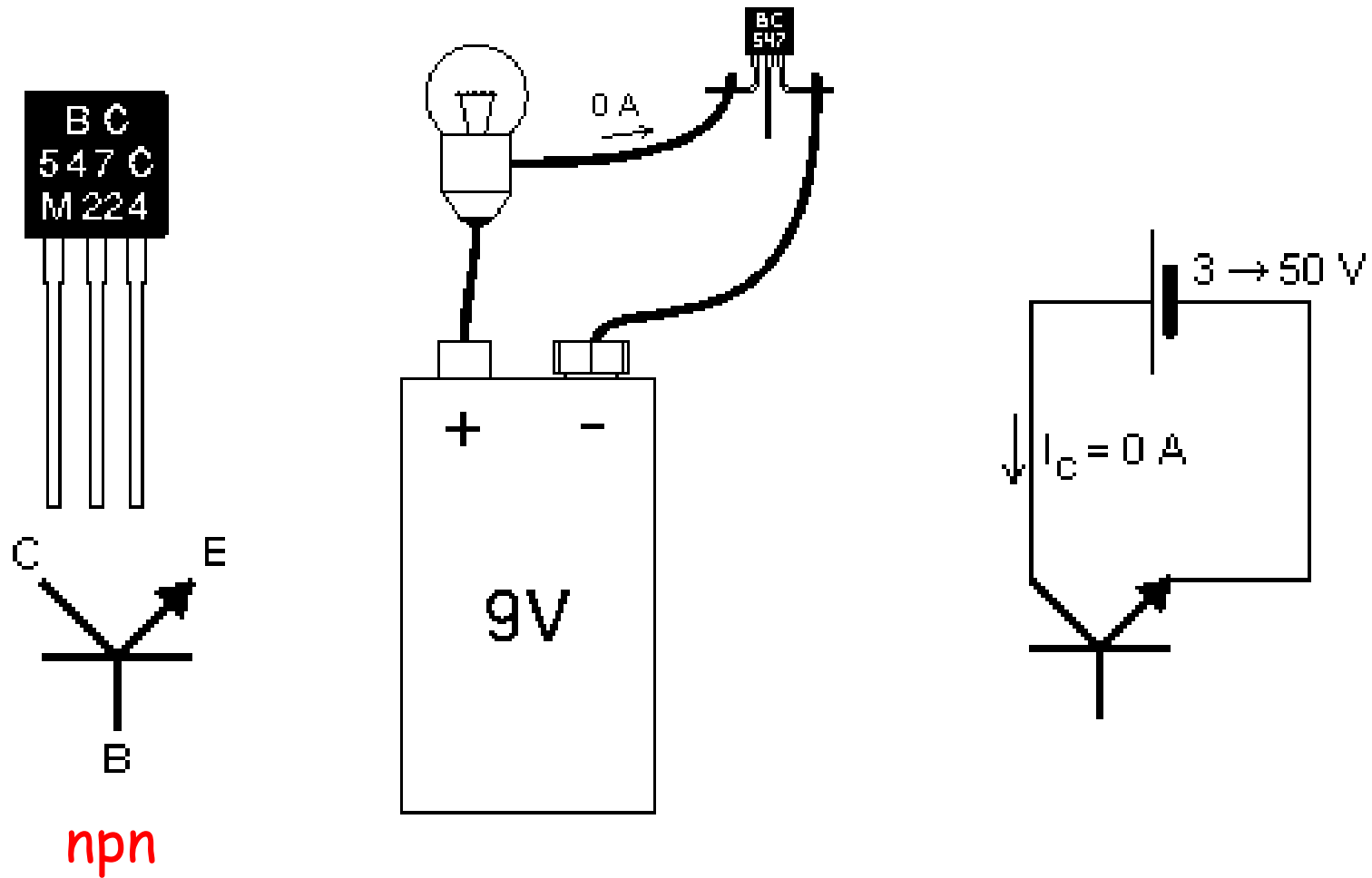
$$V_{om} = 25mA * R_L = 0.5V$$

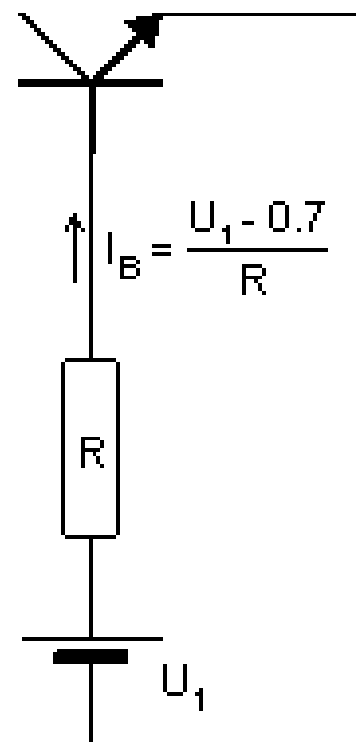
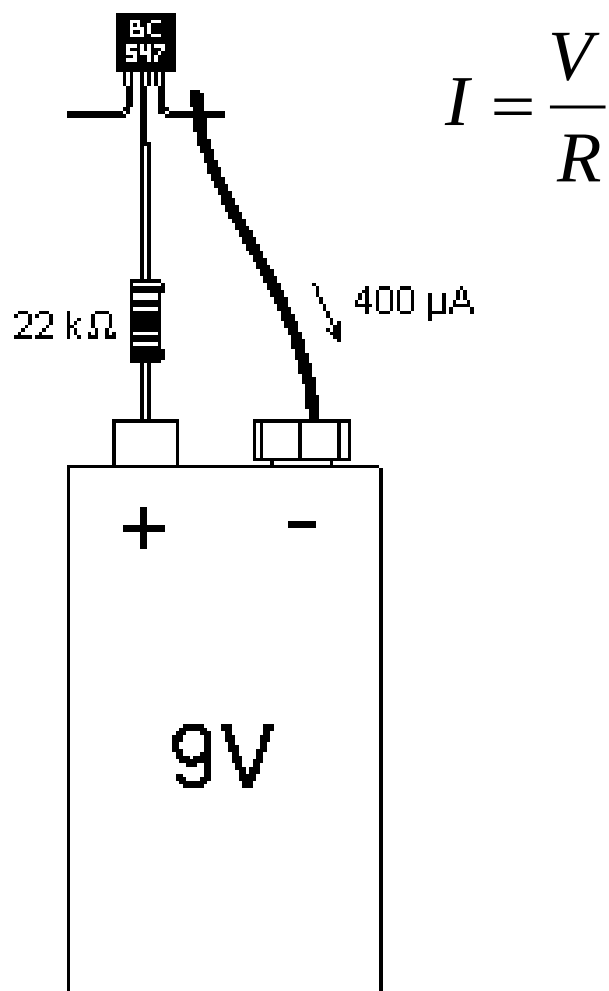
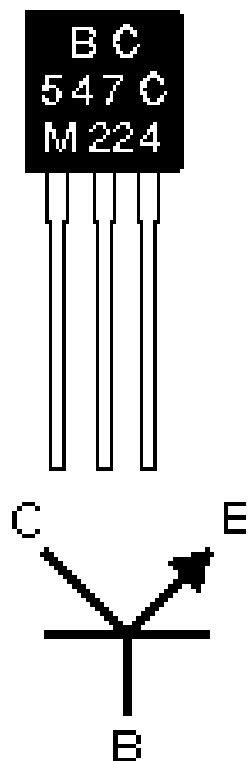
- **Case b:** Frequency of 5 kHz and $R_L = 2.5\ k\Omega$
 - **Soln.:** $V_{om} = 12\ V$ (The maximum voltage that the op-amp can achieve.)
- **Case c:** Frequency of 50 kHz and $R_L = 2.5\ k\Omega$
 - **Soln.:** The slew-rate limit of the op-amp limits the peak output voltage.

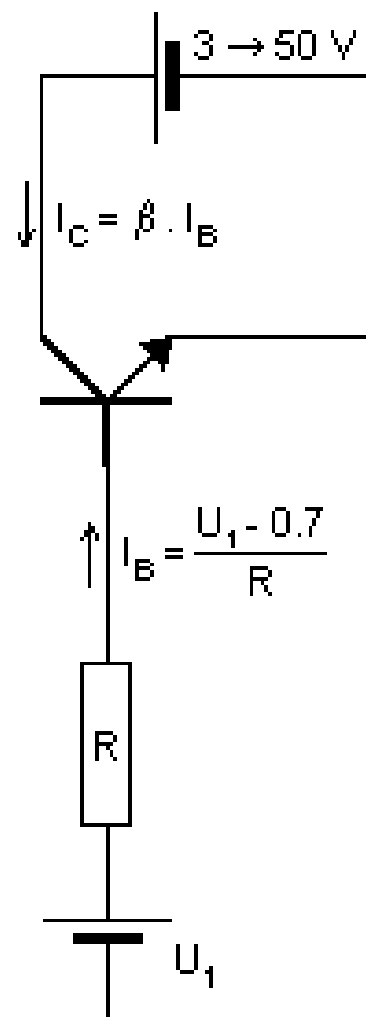
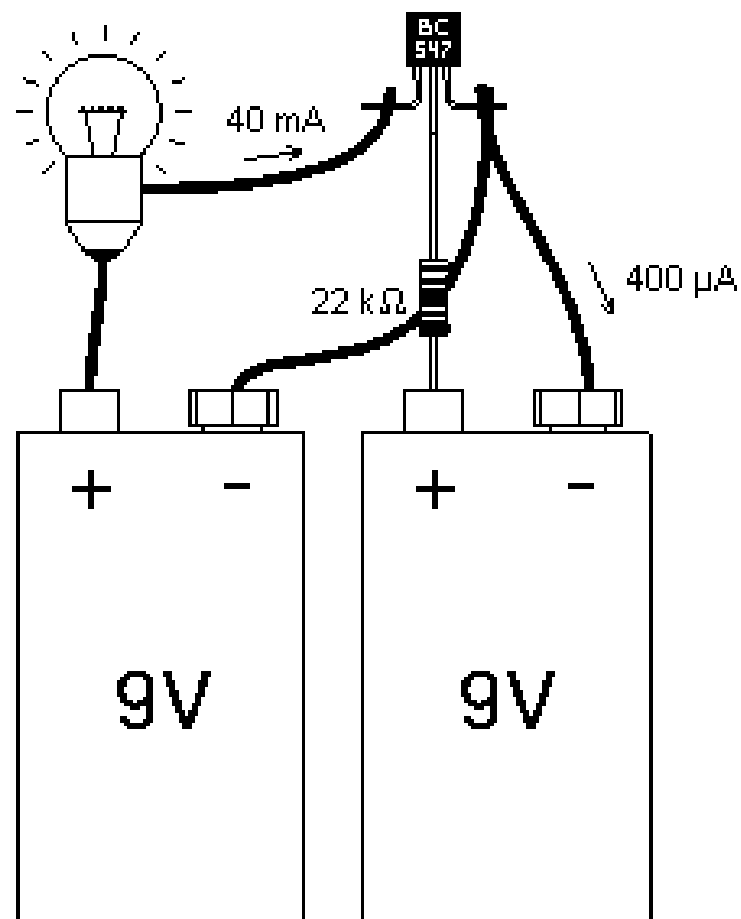
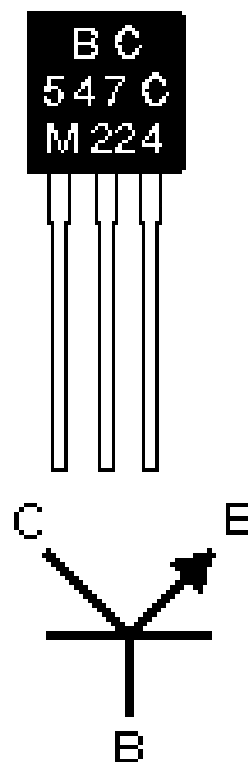
$$V_{om} = \frac{SR}{2\pi f} = \frac{1.5 * 10^6}{2\pi(50 * 10^3)} \approx 4.7V$$

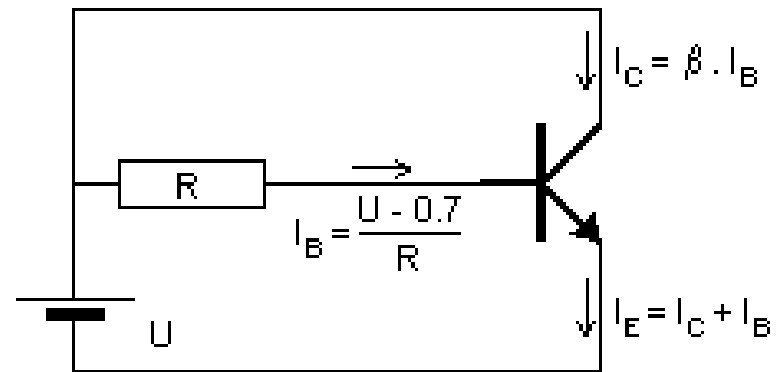
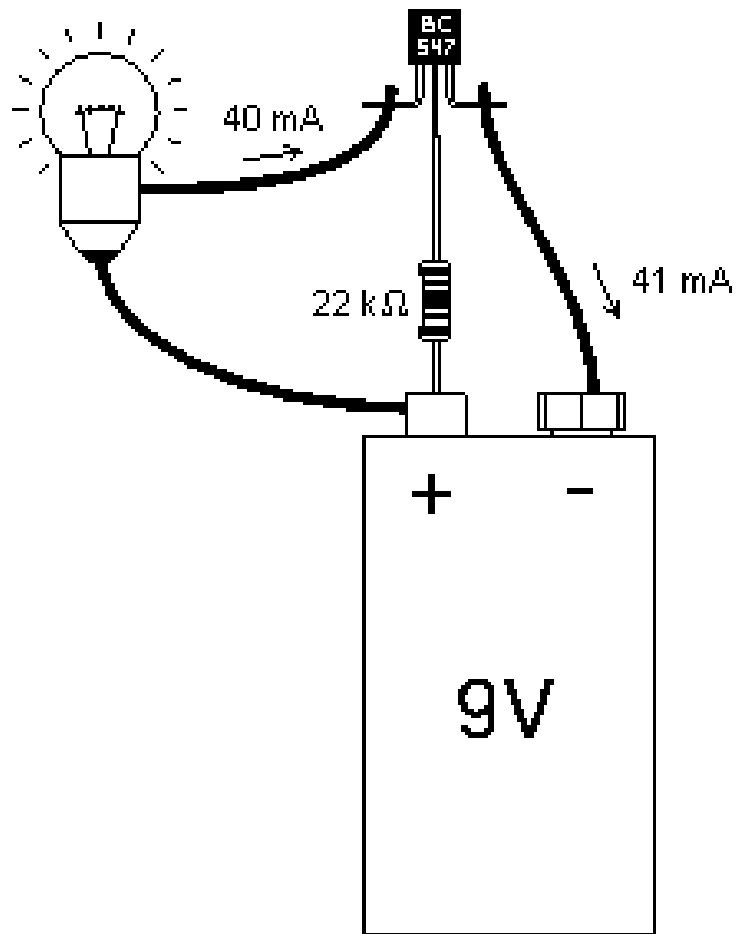


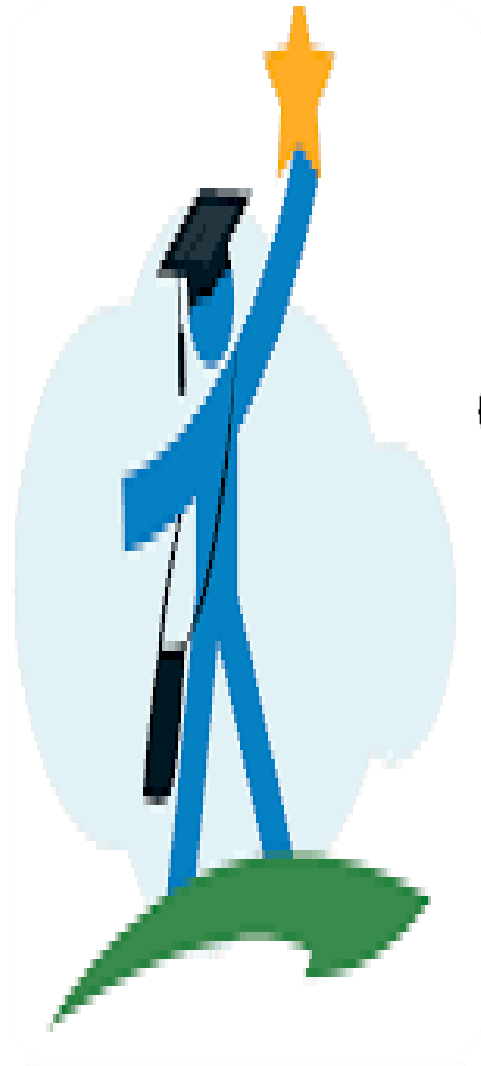
BJTs - Practical Aspects











Thanks!

